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Tattinder

Final Project Report

DT228

BSc in Computer Science

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30th March 2023

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Abstract

This project will focus on the implementation of a web app, designed to browse, and save tattoos. The web app will also have a recommendation feature which will allow the user to get recommended a tattoo based on the content of their liked tattoo. In this document I will discuss all the research and implementation carried out in order to facilitate the development of this project.

The tattoo industry has become overburdened and currently supply cannot keep up with demand, In a recent survey carried out online it was found 46% of Americans have at least one tattoo **(1)**. The unchecked demand is largely due to that fact every tattoo needs a consultation to decide on which tattoo the client would like. Of course, for custom pieces this step is required however a large majority of first-time clients do it the traditional way and pick the tattoo they like best from the display book, even if that tattoo is not necessary their ideal design.

With the use of a recommendation system this time-consuming process can be skipped over as the client would already have a design which is based on their personal preferences.

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Declaration

I hereby declare that the work described in this dissertation is, except where otherwise stated, entirely my own work and has not been submitted as an exercise for a degree at this or any other university.

Signed:

Aaron Healy

Aaron Healy

30th March 2023

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Acknowledgements

I would like to begin by expressing my sincere gratitude to my supervisor, Richard Lawlor, for his unwavering guidance, support, and encouragement throughout the course of this project. His knowledge and expertise in the field has been invaluable, and his willingness to help me overcome challenges and improve my understanding has played a significant role in my growth as a researcher.

I would also like to extend my appreciation to the faculty and staff of the School of Computer Science at Technological University Dublin for providing a conducive learning environment and access to essential resources that have facilitated the successful completion of this project. Their dedication to fostering academic excellence has been inspiring, and I am grateful for the opportunities to learn from and work alongside them.

To my fellow classmates and colleagues, thank you for the stimulating discussions, constructive feedback, and camaraderie throughout this journey. Your insights and perspectives have enriched my understanding and contributed to the development of this project. I am grateful for your companionship during late-night study sessions and for the moments of laughter that have lightened the load of academic pursuits.

I would also like to acknowledge the support of my friends and family, whose unwavering belief in my abilities has been a constant source of strength and motivation. Your understanding, patience, and encouragement have been essential in helping me maintain a healthy work-life balance, and I am deeply grateful for your love and support.

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1. Introduction

This section will cover the project background, the aims and objectives of the project, a brief description of the project and a roadmap of the implementation.

1.1. Project Background

Tattooing is the art of inserting ink into the top layer of skin by perforating skin with a tattoo machine. There are many different reasons a person may want a tattoo, however one of the most difficult things about the process is picking which tattoo you would like as there can be many unknown factors, how the tattoo will look when finished, the healing process, will the tattoo fade or lose definition over time. Every single factor must be taken into account when considering a tattoo as it is permanent, many people can feel conflicted or unsure about picking a first tattoo. However, with the use of modern technology and software architectural methods it is possible to solve part of this issue.

Tattoos are a growing trend in our society and in the last generation alone have skyrocketed in popularity. More and more people every day are seeking out artists in a bid to get there first tattoo, whether it be a deceased family member, the lyrics to their favourite song or even a wedding ring. There are many different styles and categories that a tattoo may fall under, however when starting out it can be very difficult to choose a first tattoo.

1.1.2 Tattoo History

Even though tattoos are seen as a relatively new, the length of their history can be surprising with the oldest tattoo equipment being tools found that were dated to be 2700 years old(2), these tools were sharpened bones used to prick the skin and rub soot into the open wounds to permanently change the skin pigmentation.

The word tattoo comes from the sound the old instruments used to make tattoo would emit, originally this sound was a hollow tapping as the instruments used were bones, over time the name evolved into what is commonly used today “tattoo”. The oldest physical evidence of a

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tattoo would be Ötzi (3), A mummified human who lived somewhere between 3350 and 3105 BC. His body has been preserved in a glacier on the border of the Austrian and Italian alps, he had 61 tattoos in total and is currently on display in a museum in southern Italy.

Tattoos are also known to have been used in ancient Egypt, religious people often used tattoos as a way of bringing themselves closer with the gods. However, Egyptians priestesses also used tattoos in the form of collections of dots and lines, as they thought it increased fertility(4). After this, tattoos became more synonymous throughout society with people in both the Roman and Greek empires using tattoos to indicate ownership, military status, criminality, profession or for religious purposes.

With the rise of Christianity tattoos popularity plummeted as they were frowned upon by the church, this meant that tattoos in the western world had almost become a thing of the past. The downwards trend continued and in 1891, Samuel O'Reilly invented the tattoo machine(5), based on Thomas Edisons electric pen(6), which could perforate the skin up to 50 times per second. In contrast before this invention artists could perforate the skin with traditional tools roughly two to three times per second. Considering O'Reilly's invention, the speed of tattooing was now increased by roughly 2000-2500%.

In the 1960's tattooing was banned in New York City as the government feared a potential health crisis through spreadable blood diseases, this ban lasted for nearly four decades and tattooing was eventually made legal again in New York in 1997(7). By the time tattooing became legal again in New York, Christianity has somewhat lost popularity in the western world and with this people began to see tattoos as inspirational and a form of self-embodiment.

1.1.3 Modern Tattooing

Even though the history of the craft is undeniable it is clear tattooing in recent years has exploded in popularity, with recent surveys finding Italians favour tattoos the most of any nationality, with 48% of Italians having at least one tattoo. This statistic is closely followed by Sweden at 47% and the United States with 46%(8).

There is also statistics regarding preferences in tattoos between genders. Even though the number of women who get tattoos is higher than men they are also more likely to get that tattoo removed, a male is statistically more likely to get the disliked tattoo covered up with another one. Interestingly, Chinese, and Japanese language characters are most likely to be used as inspiration for tattoo designs of any language and are most common in the United States where people translate a meaningful quote to either Chinese or Japanese and then get the phrase tattooed in the foreign characters. Tattooing is now the sixth fastest growing industry in the United States and generated an estimated 1.6 billion dollars in 2022(1).

1.2. Project Description

This project will consist of a three-tier architectural system in the form of a webapp. The webapp will be simplistic and user-friendly, after an account has been made a user can login and start saving tattoos to their gallery through a search function. A user can also navigate to the recommendation tab, where they will be given recommendations for tattoos that are deemed most similar to their previously rated tattoos by the algorithm.

Due to increased demand, many tattoo artists are struggling to schedule both consultations and appointments to get tattooed. Consultations can occasionally take up to one hour per client and all problems are sometimes not solved within that one hour. Therefore, a significant portion of the artist's time is spent on steps in the process that don't call for special knowledge. By adopting a recommendation system users can get recommended a tattoo based on their interests without them having to squeeze in time to accommodate a consultation. This can also aid those looking to get their first tattoo by providing them at least a general notion of the kind of tattoos they tend to appreciate.

Many extremely well-known internet services, like YouTube, Netflix, Spotify, and the majority of online retail, commonly use recommendation systems. It is possible to tackle the problem of facilitating consultation times and providing users with a platform to receive recommended tattoos based on interest and similarities between the tattoos by developing a web app which recommends users tattoos.

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1.3. Project Aims and Objectives

The overall aim of this project is to build a webapp that will facilitate the storage of tattoo images liked by the user, the webapp will also offer the user the option to get recommended a tattoo based on the ones they rated previously. This will be achieved by implementing a user interface in the form of a webapp, which will save user preferences and give users recommendations based on similarities measured in back-end development.

- Overall objective of this project is to give the user a simplistic webapp for browsing tattoos.
- Webapp will also feature a recommendations tab.
- A scalable user-friendly front-end implementation will be used for both login/register and user navigation.
- Back-end implementation will feature a means of storage to store relevant information regarding each tattoo and the logic needed to develop the website.

1.4. Thesis Roadmap

2. Literature Review – A brief overview of existing solutions and research completed in domain.
3. System Design – Architecture of webapp.
4. Software Development – A discussion outlining the software development process.
5. Testing and Evaluation – Planned incremental testing of project and how progress was evaluated.
6. Issues and Future work – Risk mitigation and possible future changes.

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2. Literature Review

2.1. Introduction

The following section expands on the research done on existing solutions to recommender systems and the various ways they can be implemented, technologies that were considered for implementation of the project are also expanded upon.

2.2. Alternative Existing Solutions to Your Problem

Netflix is one of the biggest and most recognizable online multimedia platforms in the world, Netflix provides users with a large number of television series and movies, much of which are their own productions. Netflix popularity is partly in fact due to their recommendation system which offers recommendations based on what Netflix's systems deems to be the most similar match to other content in which the user has viewed. It is estimated over 75% of Netflix's views come from user recommendations (9).

Amazon is inarguable the biggest online retailer in the world and uses recommendation systems in combination with ecommerce to attract buyers from all over the globe. Amazon uses item-to item collaborative filtering which is a common recommendation system implementation method. Amazon even has a recommendation system separate from the core algorithm, which increases the probability of a user being recommended an item of less similarity, but which is on sale. This has had huge positive benefits for Amazon as they can sell off more items during a sale to capitalize on profit however also provides users with deals, they would otherwise miss out on purely because of similarity percentages in the system.

Spotify is the most popular online music listening platform and employs a recommendation system based on user-to-user similarity metrics of historical listening data. Every week Spotify presents users with a new playlist called "discover weekly", this is a personalized playlist created by a recommendation algorithm to give the user a playlist depending on what they listened to the week previous. This is very popular amongst users as it gives them fresh music

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to listen to weekly and keeps them up to date on trending music while also catering to personal taste.

Although all the above use recommendations systems to huge success in their respective industries, none of them specifically employ a recommendation system based purely on tattoos. There is a number of online platforms which distribute and store image files of tattoos for users, however upon carrying out research no solutions could be found online for a tattoo recommendations system. The lack of an easily accessible alternate solution is part of the inspiration for this project.

2.3. Technologies researched

2.3.1. Front-End Development

Front-End Development is the processing of using various programming languages to build a GUI, graphical user interface, to allow a user to interact with the web application. The main technology which will be used for Front-end development will be React, a JavaScript library for building intuitive user interfaces. With this JavaScript library is it possible to create nearly any desired user interface for a wide variety of webapps.

Front-End development is largely seen in industry as the fastest part of any project to design and implement as the technologies used are simplistic and minimal. Performance is one of the key concerns of Front-End development as the need to ensure web applications validations and render times are acceptable is one of the most important processes of any web app project.

Front-End development is also one of the main way developers can ensure accessibility of their webapps, this is done by developers through the implementation of responsive web design in order to maintain the original website design despite client environment/browser.

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2.3.2. Back-End Development

Back-End Development is the process of using various programming languages to develop server-side technologies, in simple terms this is all the parts of a web application in which the user cannot see. There is three crucial parts of Back-End development, these are the development and implementation of the server, the implementation and logic of the app itself to allow the app to manage requests from the client and send back the relevant information from the database tier. The database is a form of back-end development however is often seen as a standalone technology because of its importance to any application. Back-End developers in industry usually focus on the maintenance, solving of bugs and implementation of new features for the web application. Back-end development is often seen in industry as the most crucial and time-consuming part of the project, if back-end is struggling to be developed the whole website may never get off the ground.

2.3.3. HTML:

HTML stands for HyperText Markup Language and is a standard markup language used in web development. HTML is one of the most popular web development languages and most beginner web developers use HTML as their first language! HTMLs main objective is to help the developer structure websites and as HTML is a markup language this is achieved using tags. HTML is also universal as it is supported by all web browsers (Chrome, Firefox, explorer) and operating systems (Windows, Linux, Macintosh). In 2021 HTML was the second most used programming language in the world (10).

Advantages:

- Lightweight and fast to load.
- HTML has huge compatibility and simplistic integration with other languages such as JavaScript and CSS.
- Code is easily readable allowing for simplistic updates as project progresses.
- Loose syntax.

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Disadvantages:

- HTML is a static language meaning it cannot produce a dynamic output by itself.
- Lots of code required in HTML to create a basic webpage.
- Very limited security features with just HTML.
- Creating tables and forms in HTML can be quite time consuming.

2.3.3. CSS:

CSS stands for Cascading Style Sheets, CSS is a technology used to change how and where HTML elements are displayed to the user on a webpage. CSS can also format multiple webpages at once by using an external .CSS file to move and alter content contained within whichever markup language was used (HTML, XML). In 2021 CSS was joint second most used programming language in the world (10).

Advantages:

- Improves site speed as very little code is required to implement changes.
- Less complex than other formatting options which means lots of time saved.
- Responsive web design means the webpage can be accessed by much more devices.
- Allows the developer increased control of element positions and formatting.

Disadvantages:

- Many CSS versions are used throughout the internet, this can often lead to confusion for developers.
- Security options are limited.
- CSS is not cross browser compatible which can lead to issues with so many devices on the market.

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2.3.3. Python:

Python is a high-level, dynamically typed language which is object-oriented. It was developed by Guide van Rossum and first released on February 20th, 1991 (Source). Python is one of the most used languages today and is often used in the development of websites, automated tasks, data analysis and software in general. In 2021 Python was the third most used programming language in the world (10). Python can be used for a large variety of issue and is not seen in industry has having one specific use, this sounds not ideal however it means the developer has more fundamental control and creativity with whatever is required to be implemented.

Advantages:

- Python is a high-level language means fewer lines of code are required to complete tasks, that would otherwise require more lines of code in low level languages such as Java and C.
- Interpreted language means at runtime python executes code line by line. When an error occurs, the developer will know the exact line as the code stops executing.
- Python is portable, this means the same code can be used on different machines and the same results will be achieved.
- Python has a large number of libraries available to the developer meaning most functions can be handled by the language itself without the use of external libraries, if a library needs to me imported however python also makes this easy for the developer through the Python package index.

Disadvantages:

- Python has poor memory efficiency, meaning as a program gets bigger memory management gets problematic for the developer.
- Python executes code one line at a time, this in turn means python is slow to run. When working on a project where speed is an essential factor this should be considered.
- Runtime errors often occur in python as the datatypes can change suddenly in any dynamically typed language.

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- Python does also cannot use block comments, this makes it hard for developers to preserve clarity in a large program.

2.3.3. Java:

Java is an object-oriented programming language developed by James Gosling in 1995(11). There are 9 million Java developers worldwide and approximately 3 billion mobile phones also run Java (12). Java allows the developer to code a program on one platform and simultaneously run it in another environment using an object-oriented approach. In turn this gives Java enormous reach in the industry as the use of cross platform applications can greatly impact on the cost of production for any company/developer.

Although Java is just over 2 decades old it remains an up to date and reliable programming language in the modern IT world. Java has been seamlessly developed by devoted Java developers from throughout the community as and a result Java has shaped much of the technology developers use today.

Advantages:

- Java is an object-oriented programming language meaning it is mainly used to program solutions based on object manipulation through the use of functions, classes, and methods.
- Java can run on almost any machine and doesn't not depend on any external software to be able to run, however the java virtual machine does need to be present on the machine.
- Distributed approaches – This means Java can be used to perform a number of tasks across various machines on the same network, this gives Java the ability to design a program for a distributed network while also concentrating on data and complexity.
- Security – as Java uses classes the developer can choose which classes are accessible and from what other classes/function.

Disadvantages:

- Java is very memory consuming as it is not a natively compiled language. Java source code needs to first be interpreted by the machine before it can be run.

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- Java is also quite slow because of it not being a native language.
- As Java speed and memory consumption is on the bad side this in turn drives up the price of implementation in any company/development cycle.
- Readability of Java code is slightly lacking, on a smaller program Java readability could be better however as a program expands readability declines because of Java functions and methods being quite long semantically.
- Low-level programming is not present in Java, an example of this is the fact pointers cannot be used.

2.3.3. Node.js:

Node.js is a JavaScript framework back-end runtime environment engine. Node.js is mainly used by developers worldwide for server-side functionality programming, where the language focuses on events. Ryan Dahl developed Node.js in 2009 and it was designed primarily for building fast and scalable network application. As Node.js focuses on events it is lightweight and efficient, also meaning Node.js is low in cost which makes it extremely popular across the industry while also catering to data related applications and distributed systems.

Advantages:

- The main advantage of Node.js is its speed. This is a result of Node.js being based on Chrome's JavaScript engine which is extremely fast in the execution of code.
- Node.js is a single threaded and as a result can handle a much larger volume of requests than a traditional multi-threaded server. This is because of the event driven nature of Node.js and how it concurrently handles requests.
- Node.js does not buffer output of data which means all JavaScript code run in Node.js will be outputted in chunks of data.
- Node.js allows for JavaScript to be run on both the client and server side, allowing for developers to only use a less extensive range of technologies during application development.

Disadvantages:

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- Node.js runtime programs can get very complicated when used in combination with a relational database.
- Has a very steep learning curve if the developer is not already proficient in JavaScript.
- Asynchronous nature of programming model makes it difficult to maintain and document a large development process using Node.js
- API for Node.js is notoriously unstable and frequently has many bugs.

2.3.3. JavaScript:

JavaScript is a programming technology used on the web which is the most used programming language is the world and used by 97% of all websites worldwide (12). JavaScript serves as one of the core technologies that make up the world wide web. Before this only HTML and CSS were used to develop websites, however with the development of JavaScript developers began to be able to design and implement interactive websites, some of the most popular and well-known websites in the world are built on JavaScript including Facebook, YouTube, and Google.

JavaScript also forms the basis for many extremely popular frameworks which include React, React native and Node.js. Frameworks are libraries of JavaScript code developed and distributed to other developers in order to simplify coding when implementing common features. JavaScript is also widely used in developing games that are rendered and played in the user's browser. One of the most recognizable and infamous games developed originally in JavaScript is Angry Birds, which now even has its Hollywood movie production.

Advantages:

- JavaScript is an interpreted language which means it is quite fast, this is because unlike other languages like Java and Python the machine does not need to translate code before running it.
- JavaScript will always be run on the client-side environment meaning lots of bandwidth will be saved, this also makes the execution time of JavaScript increased.
- JavaScript can be used cross-platform and does not need different code bases for viewing the same website from a different browser.

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- With the inclusion of Node.js as a communication between servers metric, the developer can make an application using only JavaScript for both front-end and back-end development.

Disadvantages:

- JavaScript's biggest fault lies in the fact that anyone loading a webpage which uses JavaScript can view the source code. This is not good as it can mean possibly security vulnerabilities in both small and large scale projects.
- JavaScript can be difficult to debug as errors can remain unclear.
- JavaScript only supports singular inheritance which can lead to issues for developers once a project starts to increase in size and complexity.
- If JavaScript code contains an error the entire webpage may stop functioning until that singular error is resolved.

2.3.3. PHP:

PHP stands for HyperText Pre-processor and is used as a server-side scripting language, it is a back-end development technology. PHP was developed by Rasmus Lerdorf in 1995 (13). PHP is used to develop dynamic and interactive websites and it was the first scripting language designed that could be embedded into HTML. The development of PHP was crucial as without the use of embedding early web developers would have struggled with adding functionality to web pages. PHP is also commonly used in the design and implementation of web servers and can even be run in the command line.

Advantages:

- Flexibility is PHPs biggest advantage, PHP can be used in coordination with a wide variety of other programming languages making it desirable to learn and use for any developer.
- PHP is a cross platform technology, this means that users can use PHP based applications from any operating system or browser while also accommodating slower internet connections.
- PHP has lots of libraries with can be imported and used as a way to solve commonly found problems and also for data representation.

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- PHP code is easy to read and as a result is easy to maintain and document as the development process progresses.

Disadvantages:

- PHP is excellent for developing small applications however if a project has several complexities which need to use other technologies to be developed, it can become hard implement it all with PHP.
- PHP is an open-source language which means it contains ASCII text files, these files are easy accessed by users and developers which can lead to possibly security vulnerabilities.
- One of the mains disadvantages of PHP is the fact it has very limited debugging options. This makes it hard for developers to find solutions for problems that would be otherwise easy to solve using other scripting languages with better debugging options.

2.3.3. C:

C is a very old program language and was originally designed by Dennis Richie in the 19070s, despite its age C retains its userbase and remains very popular today. C and Unix are always held in regard to each other as C was originally developed by Dennis Richie to design the Unix operating system. C is also one of the fastest programming languages today as it is very close to assembly code. MySQL which is one of the most popular database management systems was also first written using C. C can make so many different programs such as MySQL and UNIX because it is a general-purpose programming language. C is widely regarded as the precursor for modern programming languages and is believed to have inspired the development of languages such as JavaScript, Java, Python, C++ etc.

Advantages:

- C has a large variety of data types and operators which can be used during implementation.
- C can be used to develop just about anything as it is a general-purpose programming language.
- C has the ability to directly manipulate memory.
- C is a lot closer to assembly code than most other programming languages meaning it is inarguably the fastest programming language in existence.

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Disadvantages:

- C is a general purpose programming language, this can be seen as an advantage in terms of overall development but also means C cannot employ object oriented programming ideologies or architectures.
- C has no “safety net” so to speak, when dealing directly with memory it is very easy for a developer to make a detrimental mistake in allocation, this in turn can lead to crashes. Modern technologies and languages mostly do not deal with manual memory management as a result of this.
- C is close to assembly code which means it can be a fair bit more complex than high level languages during development, A lot more brain power has to go into exact functionality requirements and implementation in C than in object-oriented or scripting programming languages.

2.3.3. C++:

C++ has syntax very close to the programming language C. This is because C++ was originally designed as an object-oriented approach to the C language. It is viewed by many developers as the best object-oriented language to develop large-scale systems and applications. Similarly, how C++ is based on C, Java is then based on C++. C++ is also a procedural programming language, this is one of the most important features in C++ as opposed to C. As a procedural programming language C++ executes code at runtime in a structured based approach instead all at once.

Advantages:

- Cross platform development allows for extra portability in C++ as code can be run the same on Windows operating systems as Linux.
- C++ is seen as a mid-level programming language, this means it can be seen as both low level while interacting while machine kernels and drives but also seen as high level while dealing with this such as video game development and desktop/mobile applications.
- C++ is an object-oriented language meaning it can manipulate objects to employ concepts commonly used in OOP such as inheritance, polymorphism, abstraction, and

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encapsulation. This is essential as this is seen as one of the only functionalities the original language C could not do.

- Dynamic memory allocation is important in C++ as it gives developers the power to free and allocate memory as desired. In C++ the developer has complete control over memory management much the same as C.

Disadvantages:

- C++ is notoriously difficult in regard to pointers, both conceptionally and in practice. Errors at runtime regarding pointers could even possibly lead to system failure.
- C++ has various security vulnerabilities; this is a result of global variables. If one global variable is changed to the wrong type it would be easy to corrupt an entire codebase/system.
- As the programmer has full control of memory management and allocation in C++ it is easy to make a mistake and possibly increase the total memory taken for a program.

2.3.3. Flask:

Flask is a framework based on Python, used for designing and implementing web applications. It was created on the 1st of April 2010 and was originally authored by Armin Ronacher (14). Flask is commonly used by new developers as it is easy to understand, and a webpage can be created from a single Python file while using the flask framework. Flask is often used at the beginning of development cycles by developers for prototyping purposes, this is because it is simplistic and offers very fast production time to form the basis of a prototype, which can be expanded on later to add whatever the required functionalities are.

Advantages:

- Flask is easy to understand and leads to fast and simplistic development on webpages.
- Flask is minimal meaning it can be changed and altered to suite the developer's requirements without the risk of the whole programme not running.
- Flash is a very scalable framework as it is often used and deployed in cloud computing environments which makes it very easy to handle possible increased in traffic.

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Disadvantages:

- Flask is not a standardized language which means it can be very diverse from program to program. This in turn can lead to issues in documentation when developers leave and join new projects/companies.
- Flask has a lot of libraries available which means over time large projects which incorporate many libraries begin to cost more as the libraries lose support over time and maintenance needs to become more frequent.
- As flask is still seen as relatively new programming framework the community is not necessarily as big as Django for example which is another python framework.

2.3.3. Django:

Django is a high-level Python framework used in the development of web applications. Django was developed with the intention of making the nuances of web development easier allowing the programmer to focus more on the complexity of the webapp at hand. It is mainly used in combination with databases to make smart web applications.

Advantages:

- Several massive technology companies use Django including Netflix and Amazon prime for video streaming. This is because Django has very good content delivery networks, meaning clients receive multimedia content from servers in a more efficient way, leading to increased client satisfaction.
- Django can simply import libraries for a number of commonly implemented features such as auth package for Authentication functionality.
- Django has extremely good scalability as it can be run on multiple separate database servers simultaneously, this can be done through load-balancing.

Disadvantages:

- Specifications of URLs can be difficult for implement for beginners in Django.
- Is frowned upon when developing small projects as it is a high-level language, which in turn means the framework is natively catered towards more complex features.

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- Django can lead to slow websites. This is not a result of the framework itself but rather how many resources it depends upon to use.

2.3.3. Mongo dB:

MongoDB is a non-relational database technology commonly used for storage and management of JSON format files. As it is non-relational it uses NoSQL and is most often used when dealing with a large amount of distributed data. MongoDB was first released in 2007 by Dwight Merriman, Eliot Horowitz and Kevin Ryan(15). MongoDB was originally designed to give developers the freedom to focus on the developing on the application software and not having to work around the nuances of the database layer.

Advantages:

- Ease of use and implementation makes MongoDB desirable for faster production times.
- Due to the main use of document style data MongoDB offers increased speed than that of a traditional relational database.
- NoSQL means developer does need to focus on data being unstructured or structured and MongoDB can use both.
- MongoDB offers support by employed professionals, meaning no matter the bug/error it can be solved.

Disadvantages:

- Can be hard to manage effectively over time because of duplicate data.
- MongoDB often takes a huge amount of memory for storage.
- MongoDB does not support joins, this in turn can makes it difficult and time consuming to construct queries where joins are required to request data from multiple places.

2.3.3. Firebase:

Firebase is a cloud-hosted database technology which is NoSQL. It is mainly used in the development and implementation of IOS, android and webapps in industry. It was developed and released by Google and is designed to help build web applications in a fast efficient manner without the need to worry about infrastructure management. As Firebase is

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developed by Google it can be used in combination with other technologies developed and released by Google, and example of this is the use of Google Analytics which can be integrated with Firebase to give the developer insight into user actions and preferences.

Advantages:

- Large community of developers using firebase means extensive technical documentation available online.
- Non-relational NoSQL database means large collections of data can be stored while maintaining performance.
- Firebase Authentication developed by Google is designed to offer an easy to implement authentication method.
- Firebase comes with a content delivery network, this means Firebase offers guaranteed increased speed of content delivery worldwide.

2.3.3. React:

React is one of the most recognizable and widely used JavaScript libraries used for developing Front-end on a wide range of web applications and services. It was originally developed by a Facebook developer in 2011 (16). React was initially developed to mimic the functionality of XHP, which is a HTML component framework for PHP which enables the developer to reuse HTML elements, react is very similar to this however it is based on JavaScript.

React is one of the most influential programming languages in the modern day, this is because Facebook originally used it in production for their home screen feed which was used by hundreds of millions of users every day. In 2012 when Instagram was purchased by Facebook, Instagram also began to use react for its news feed functionality(16). In 2013 Facebook made the decision to make the library free to use to all developers as they believed it was crucial for the progression of the information technology industry.

Advantages:

- Experience in JavaScript makes React very quick to learn for any dev and is quite simplistic to implement most front-end requirements.
- React enables the developer to reuse code previously developed to implement a separate part of the web apps functionality.

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- Ease of testing in React is a great advantage in modern day technology.
- As React uses a virtual Dom this increases performance application wide and gives the developer more freedom in creativity as performance is always increased in contrast to traditional web app development methods.

Disadvantages:

- React is well known for having relatively lacking documentation making it hard for developers to debug certain common issues and errors.
- React can only be used to develop UI components of a webapp and thus will always need to be used in combination without another web architecture technology for back-end development purposes.
- As react is still relatively new programming language it can be hard to use on large projects as it is constantly updated and developed by Facebook and the community.

2.3.3. Angular:

Angular is a JavaScript web framework used for the design and implementation of single-page web applications by using HTM and JavaScript. Angular has a wide range of libraries which can assist developers with the implementation of common features such as form management and client-server communications. Angular uses a combination of components and templates to give a simplistic development process to the develop using angular for their single-page web applications. Angular works by diving the system architecture into 3 main parts, model, view, and controller. These three parts can be developed simultaneously or individually, as a result of this angular websites are easier to update and maintain.

Advantages:

- Angular code has a huge amount of reusability and thus saves developers time in application production.
- Angular code is easy to read which makes it fast to learn.
- Angular uses encapsulation a lot which means maintenance of code doesn't necessarily get more difficult as the application expands in size.

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Disadvantages:

- Angular cannot be used in the development of mobile devices which limits usability.
- Two-way binding inevitably makes Angular slow.
- Angular can be difficult for developers who are inexperienced in MVC to use and as a result implementation deadlines may not be met.

2.4. Other Research done

Recommendation systems have two main methods of implementation. These are widely recognized in industry and tend to be very accurate. These two methods are known as collaborative filtering approaches and content-based approaches. Each of these methods have their own advantages and disadvantages, both of which should be carefully considered when deciding which of the two methods to use, when designing and implementing a recommendation system. Many large corporations implement a hybrid of these two models to achieve optimum results when recommending a user with content, these approaches however are far too complex and data demanding to use in a webapp application of the projects scale. For this reason, collaborative-filtering and content-based methods will be researched as standalone technologies and an evaluation of which is most suitable for the project will be considered and applied.

2.4.1. Collaborative Filtering:

Collaborative filtering refers to the method of implementing a recommendation system using the prediction of what one user will like, when compared to what a user of similar nature chose. This method is by far the most used method of implementing a recommender system. Collaborative filtering in simplified words focuses on predicting what a user will rate an item based on how other users historically rated an item and analysing behaviours of the past users to accurately predict new users' preferences. The analysis of the users' preferences in regard to items is known as the user-item interactions matrix.

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Collaborative filtering itself is separated into two main areas, these areas represent the different implementation techniques used when creating a recommender system through collaborative filtering. These two methods are known as model-based approaches and memory-based approaches.

Memory based approaches consist of user-user and item-item recommender systems. The main characteristic of these two methods is that they both only use information from the user-item interaction matrix when deciding on a recommendation for a user.

User-user based approaches focus on the similarities between users' profiles and what they historically liked. This approach is known as user cantered as the recommendation is formed around the users' interactions with products and how they operate behaviourally. Item-Item based approaches focus on the similarities between items when liked by a user, for example when a user likes content, the recommendation system will offer that user content, similar to what they just liked without consulting other user's profiles.

Model based collaborative approaches rely on the computing of matrix factorisation in order to determine an accurate recommendation. These types are approaches are highly sought after for large scale applications and services because as an application gets bigger it starts to record more user data and in turn makes the matrix factorisation give more accurate results in terms of similarities when recommending content.

2.4.2 Content Based:

Content based methods rely on machine learning algorithms which is a part of artificial intelligence. This method increases the accuracy of the recommendation system by analysing data inputted by users over time. In this method the similarities are calculated based on attributes referring to the items which will be recommended, hence the name content based. Content-based systems have two main approaches, vector spacing method and classification method.

Vector spacing methods work by asking the user to input rating on a number of items, after this the recommendation system ranks what the user likes vs doesn't like and begins to calculate which other item is closest to the ones the user ranked high. With minimal inputs

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the recommendation system will output similar results, however as the user rates more and more items different genres and nuances will begin to be recommended.

Classification methods rely on making the recommendation system make decisions based on the item, which was just rated, for example if the recommendation was recommending movies, the movie title would first be considered. If there are no similar titles i.e. prequels or sequels, the recommendation system will move to decide on the next classification which would be the movie genre. This decision making process will be carried out by the recommendations system until enough classifications have been considered and a recommendation can be made to the user.

2.5. Past Projects

First Project Reviewed:

Title:

Event management System Using Image Authentication

Student:

Eamon Tang

Fully implemented android app to manage events and allows users to easily authenticate when checking into a venue/event. To design a visual authentication feature for event tickets to extract and verify authenticity through server verification.

In my opinion the image being verified through a server authentication is the most complex part of this project.

Java, XML, Ubuntu, Docket, Python, SQL were all used while developing this software.

From what can be observed seen in the dissertation document it is clear to see this was a very well developed and implemented piece of research. Overall, it can be said all the

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technology used and the reasons for using that specific tech was well backed up, one possible improvement would be to try make the page less cluttered, pages with lots of information can often be confusing for first time users and leads them to avoid using the web application again in the future, sometimes simple is better especially for android applications.

Second Project Reviewed:

Title: Image selection based on optimal characteristic analysis

Student: Jameel Briones

Sorts uploaded images based on quality of the picture when it is taken. This is done by using machine learning and image processing to filter through and rank images that are higher quality than others.

Upon carefully inspecting this project it is clear the image comparison algorithms and the feature extraction and detection complexities are the most impressive in this project, these both form a vital part at the core of the project and without them it would not work.

Python for image processing, server-side technologies, python web frameworks were all used throughout development.

Overall, this project was a very strong example of what a student can develop when they really put time and effort into it. One possible improvement would be with the image comparison algorithm, although it is very complex and impressive, at some point through testing a developer would find it may become inconsistent as the project scale increases, however overall, I would say the project is a very good piece of work.

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2.6. Conclusions

In conclusion, the technologies used to implement this project include React for front-end development and logic programming, and Firebase for back-end database management. React, a JavaScript library, was chosen for front-end development due to its widespread use, efficiency, and the abundance of resources and guides available online. This choice allows the developer to allocate more time and focus on the back-end development of the project.

Reacts ease of use and component-based structure also promotes faster development and maintainability, making it an excellent choice for creating the user interface and handling application logic.

Firebase was chosen as the back-end database for this project due to its real-time capabilities, easy setup, and time-saving libraries for handling common tasks such as user authentication. By utilizing Firebase, the developer can streamline the development process and efficiently manage data storage and retrieval.

Overall, the chosen technologies of React and Firebase provide a powerful combination that enables efficient and scalable development. The advantages of these technologies outweigh their drawbacks and align well with the project's desired outcome, making them suitable choices for this application.

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3. System Design

3.1. Introduction

In this section software methodology and system design will be explained and an overview of the systems functionality using diagrams will be given. This introduction will also include various mock-up screen for the application.

3.1.1. Mock-up screens

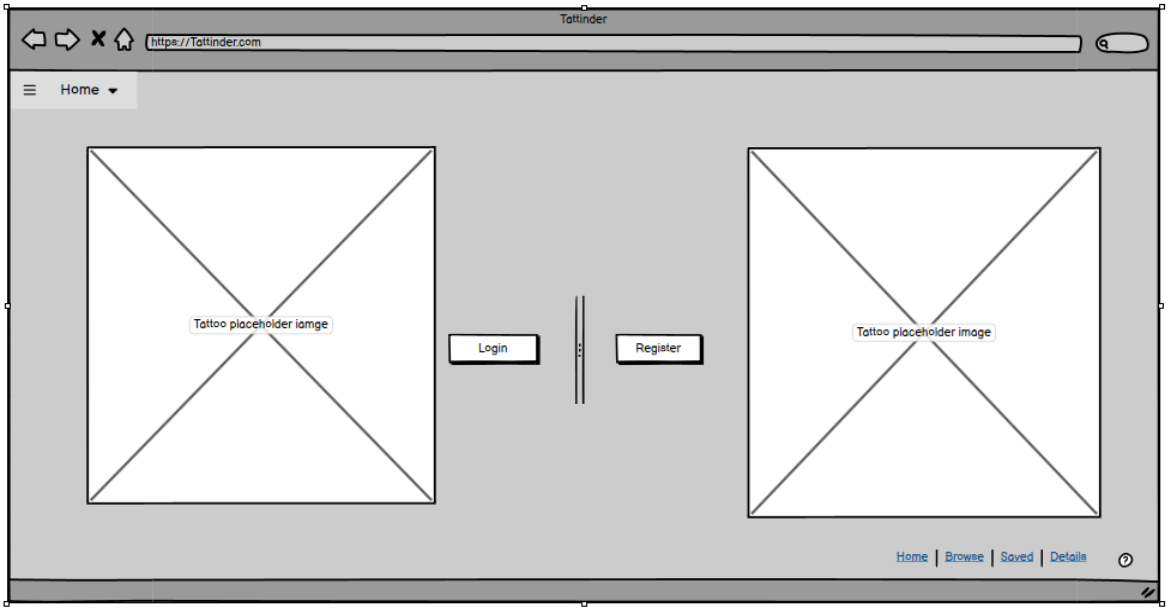

Figure 1 - Home Page

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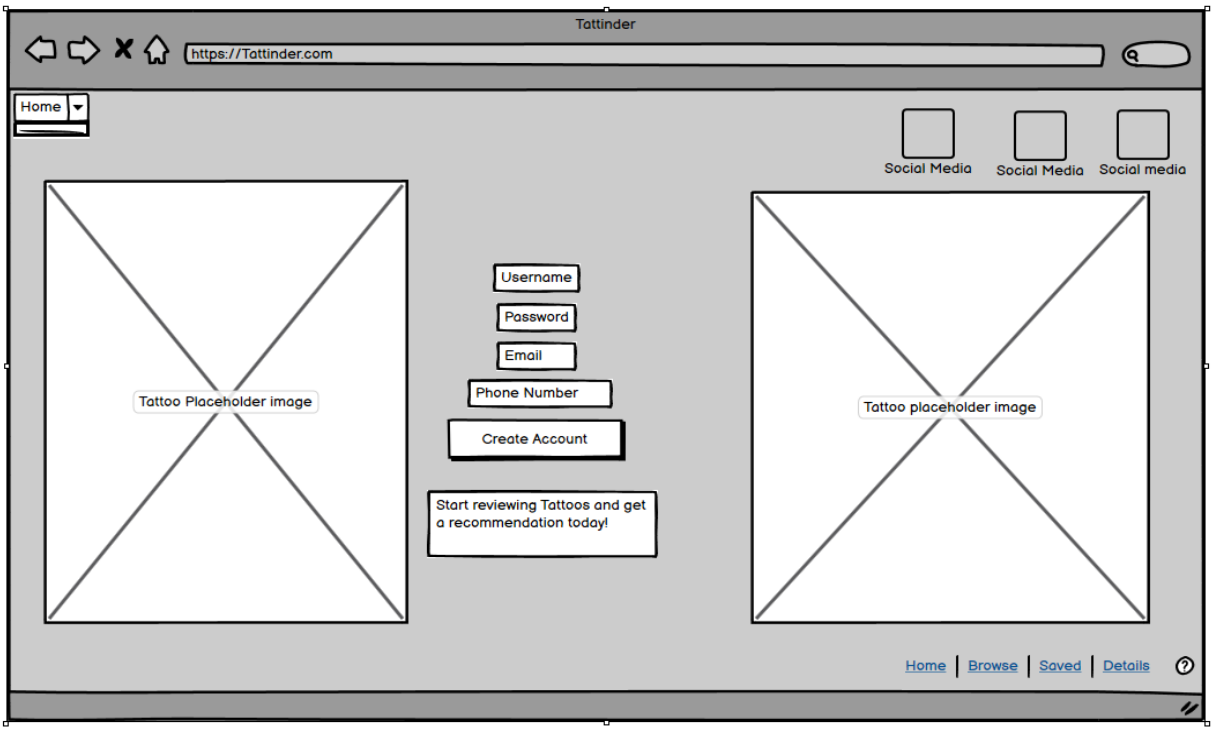

Figure 2 - Register Page

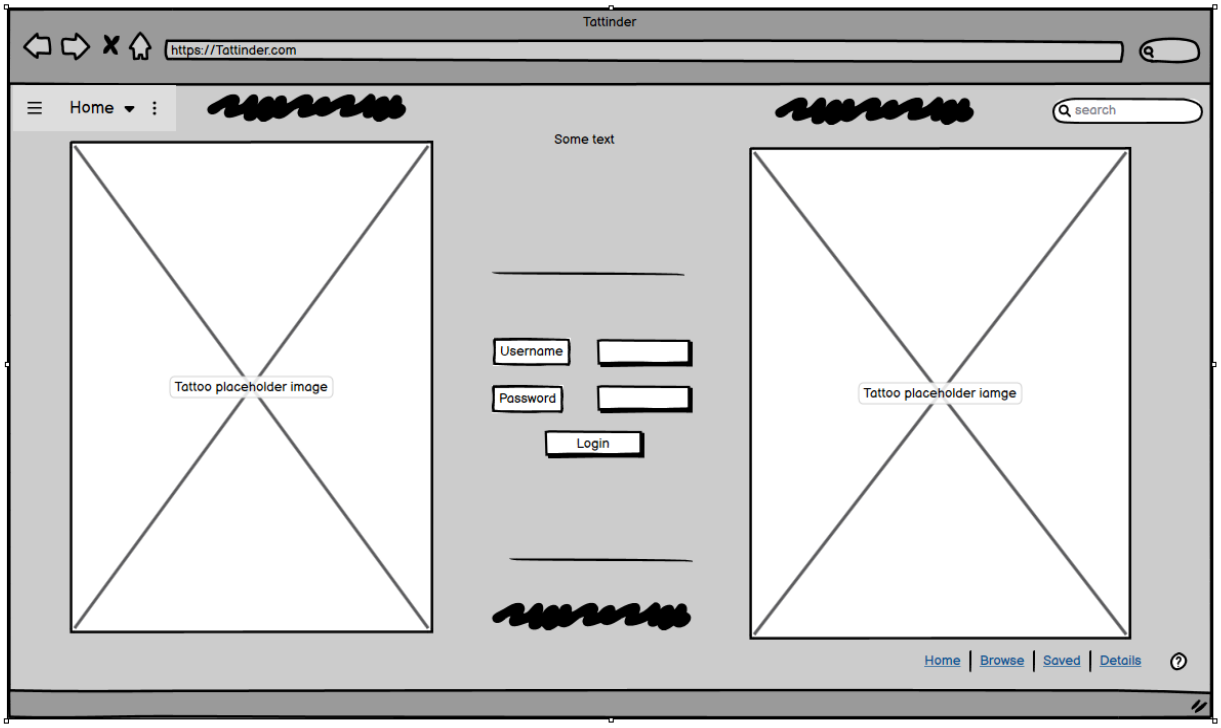

Figure 3- Login Page

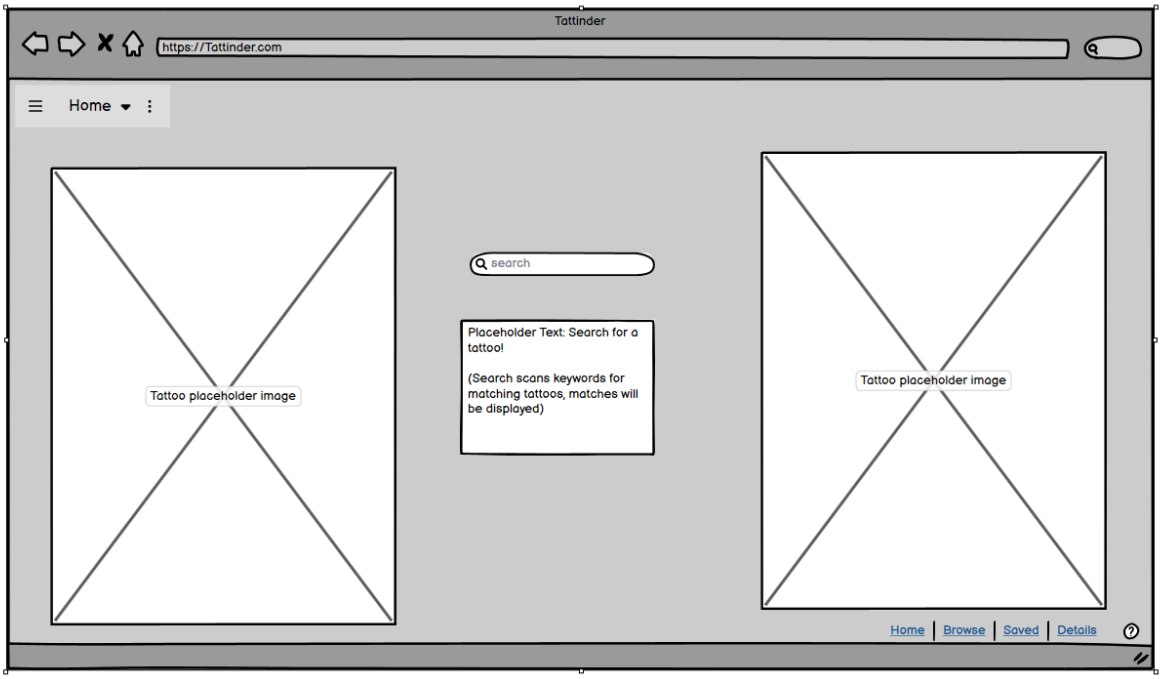

Figure 4 - Search Page

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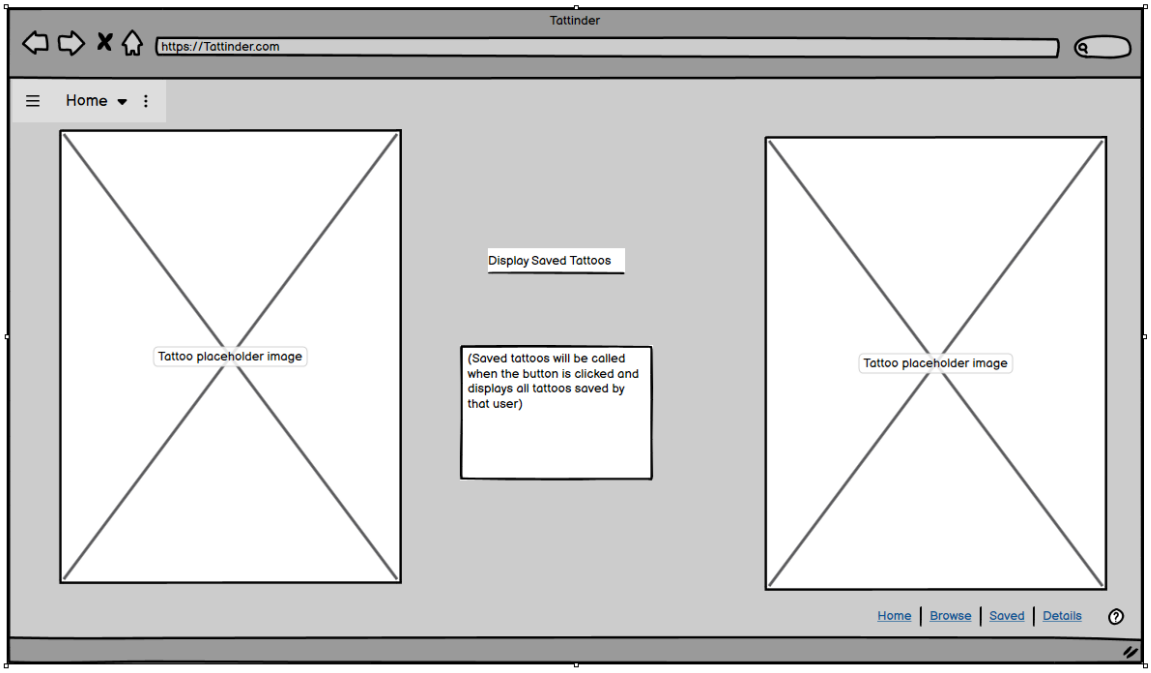

Figure 5 - Saved Page

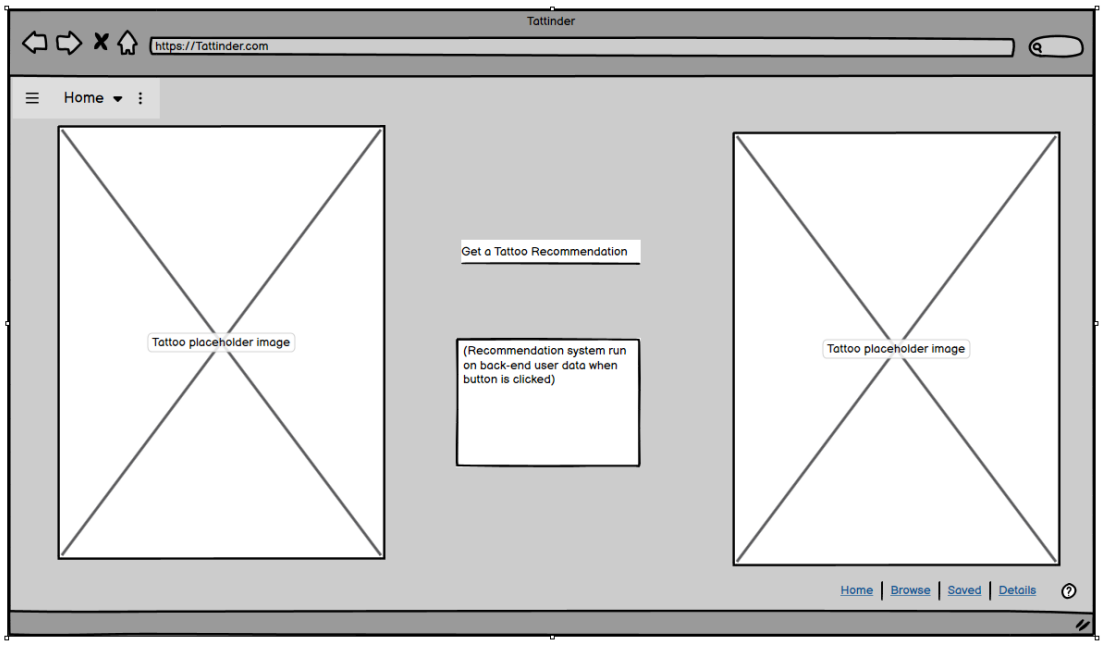

Figure 6 - Recommendation Page

3.2. Software Methodology

Software methodology refers to the structure the developer decides upon when designing a project. Software methodologies are used in industry because they give teams of developer's clear definite deadlines for stages of implementation. There are several different regularly used software methodologies circulating the programming world today. The two most popular methodologies are waterfall and agile.

Waterfall is the process of software development where requirements are clearly defined by the clients and developers can work on the project in a sequential manner. Waterfall was designed by Winston Royce in 1970(17) and remains one of the most used software methodologies to date. Projects developed using waterfall are often separated into five categories which need to be completed in order. The first category to be implemented is the projects requirements.

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Developers work on defining the requirements and once completed move onto to making a design which follows closely the requirements previously specified. Once the design has been created the project will begin to be implemented and vigorously tested. After the project has been fully implemented and tested the team of developers will then move to deploy the project and then apply regular maintenance so the project adheres to standards set out by the client.

Agile is the process of software development where requirements are given, and the developers then work continuously on each section at the same time until the project is completed. This process is usually used when the requirements are somewhat ambiguous and the need for reoccurring contact with clients throughout the development process is key. One of the most popularly used parts of agile is known as scrum. The two are very similar in nature, however scrum focuses on the overall project development and cohesion rather than strictly software requirements.

For the reasons mentioned and considered above scrum was chosen as the software methodology which was used during the implementation of this project. An overview of the scrum software methodology can be seen below in figure 7 (18).

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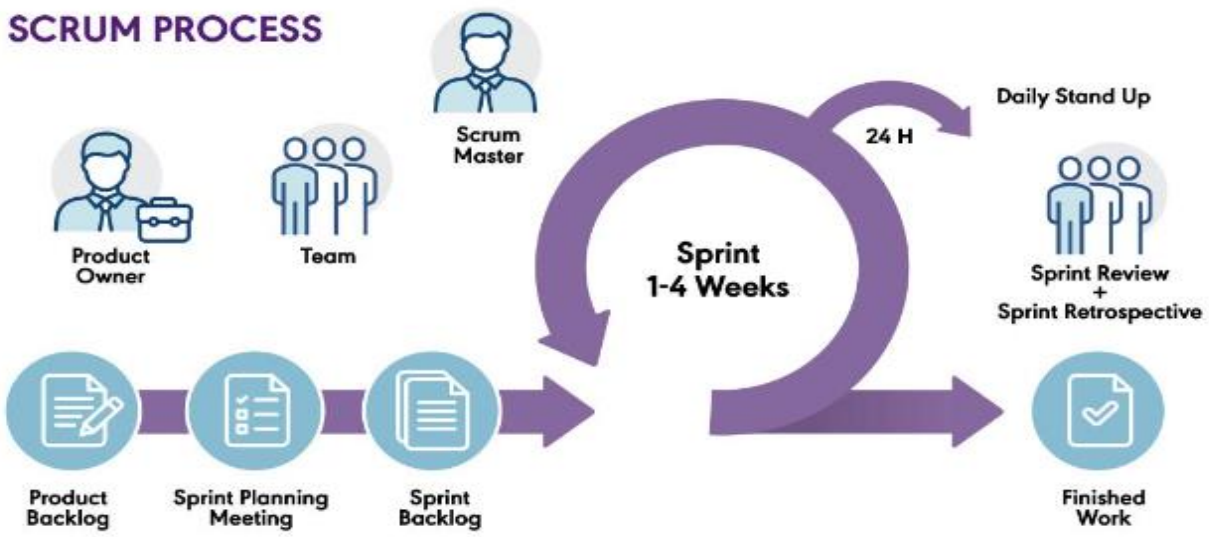

Figure 7 - Scrum Software Methodology (18)

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3.3. Overview of System

In this project, a three-tier architectural system was implemented as this was the most appropriate architectural pattern for the project. The three main architectures that made up the three tiers were as follows: the presentation tier, which encompassed all front-end technologies and methods used to create the project's front-end. The logic tier represented all methods and functions implemented in React for both front-end and back-end development, including application logic and the recommendation system. Finally, the last

tier represented the database, which stored both user and tattoo data for use in various functions throughout the project.

In this project, a combination of React for front-end, React for back-end and the recommendation system, and a non-relational NoSQL Firebase database was used. These three technologies worked excellently together and were ideal for developing a project of this scale.

React was used for front-end implementation, which was everything the user saw, commonly referred to as the graphical user interface or GUI for short. This React GUI handled all requests made by the user and subsequently sent the request to either the React-based logic or to the database, depending on the request made by the user. Finally, when the user requested to get a recommendation made by the system, the React-based recommender script was run for that user, and the appropriate recommendations were made according to their historical rating data.

A basic overview of how the projects logic system operates is as follows: The user will take an action on the webpage, depending on what action they would like to perform. From this the server will receive the users request and begins to process it. The web application will then perform whichever action the user is trying to achieve, and it is displayed on the webpage for the user to see.

Some of the use cases involved in this project include login/register, a user searching for a tattoo and saving that tattoo, giving a rating to displayed tattoos and getting content-based tattoo recommendation. Please refer to Figure 8 for a visual representation of the various use cases involved in this project.

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3.3.1 Use Case Diagram

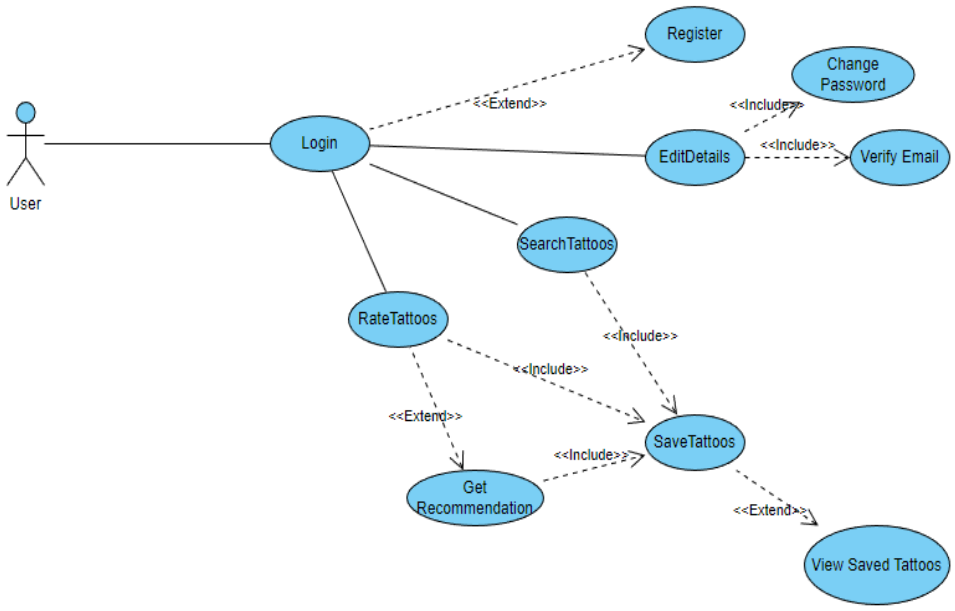

Figure 8 - Use case diagram

In terms of flow of the website, multiply sequence diagrams were illustrated to give a visual representation of the messages/requests sent between objects in the project. The sequence diagrams included represent how the project carries out tasks sequentially when the user declares a task to be carried out.

The tasks that have been illustrated to represent the tasks sequential functionality are as follows:

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3.3.2. Figure 9 – User Login sequence diagram

- The sequence used in order to complete this task starts with the user entering their details into the webpage form.
- The user's details are then checked against details in the database to verify the user has a valid account matching the details entered.
- If the entered details are valid the user is logged in and will be informed through an alert box.
- If the validation of the users entered details does not match an account in the system, the user will be asked to amend their login details.

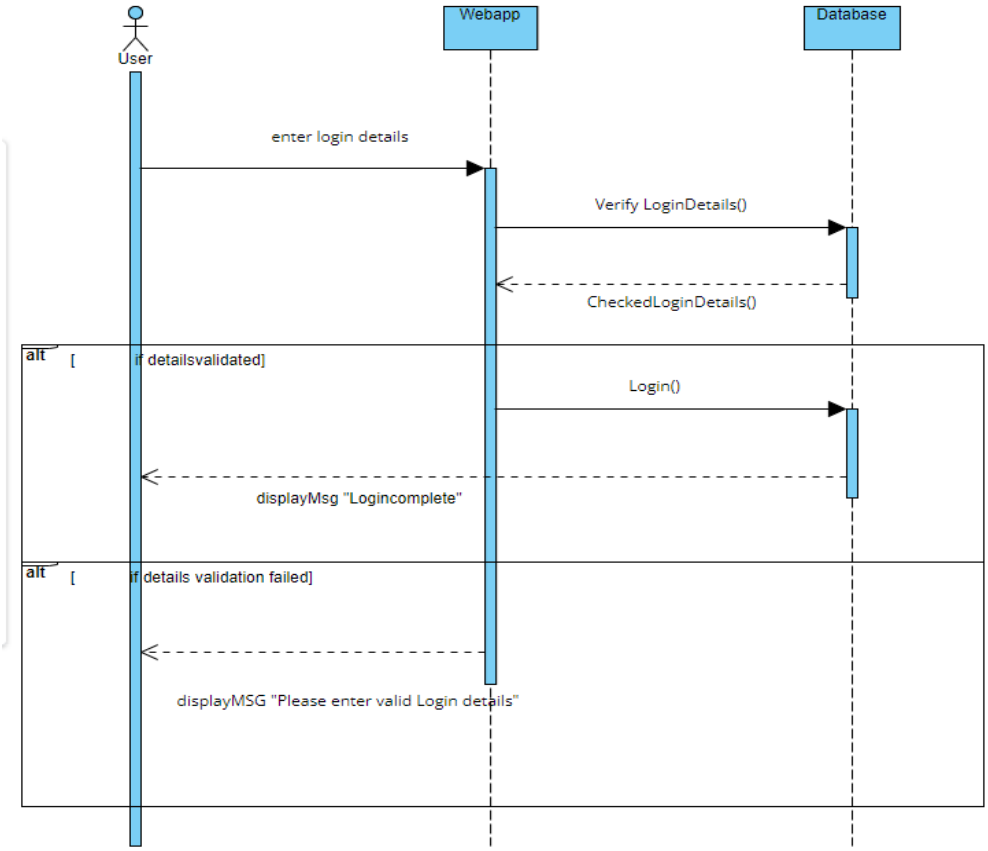

Figure 9 - User login sequence diagram

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3.3.3. Figure 10 – User Registration sequence diagram

- The registration tasks will be carried out in this project first by having the user enter registration details to create an account.
- The registration details will be validated to make sure the account can be successfully created, if the details entered match an account already in the system for example the user will be informed that username has been taken.
- If the details are valid the account will be created, and the user informed of this.
- Likewise, if the details are invalid and an account cannot be created with them the user will be asked to enter different registration details.

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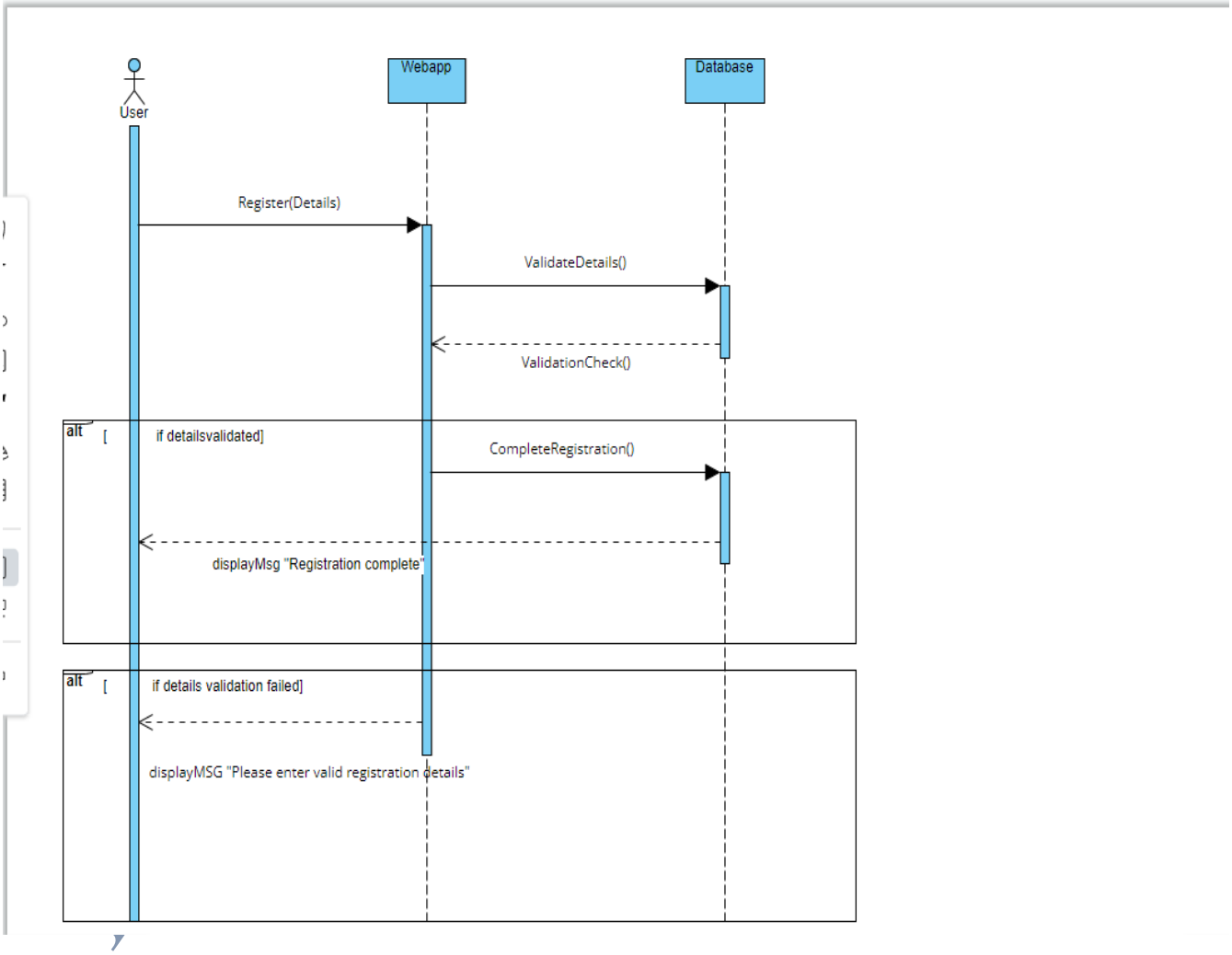

Figure 10 - User registration sequence diagram

3.3.4. Figure11 – Search function sequence diagram

- The search function will begin when the user searches for an item.
- The details of the search such as keywords will then be checked across the database to try finding a successful search.
- If the search was a success the results of the search will be displayed to the user and likewise is there is no search results the user will be asked to amend their search details.

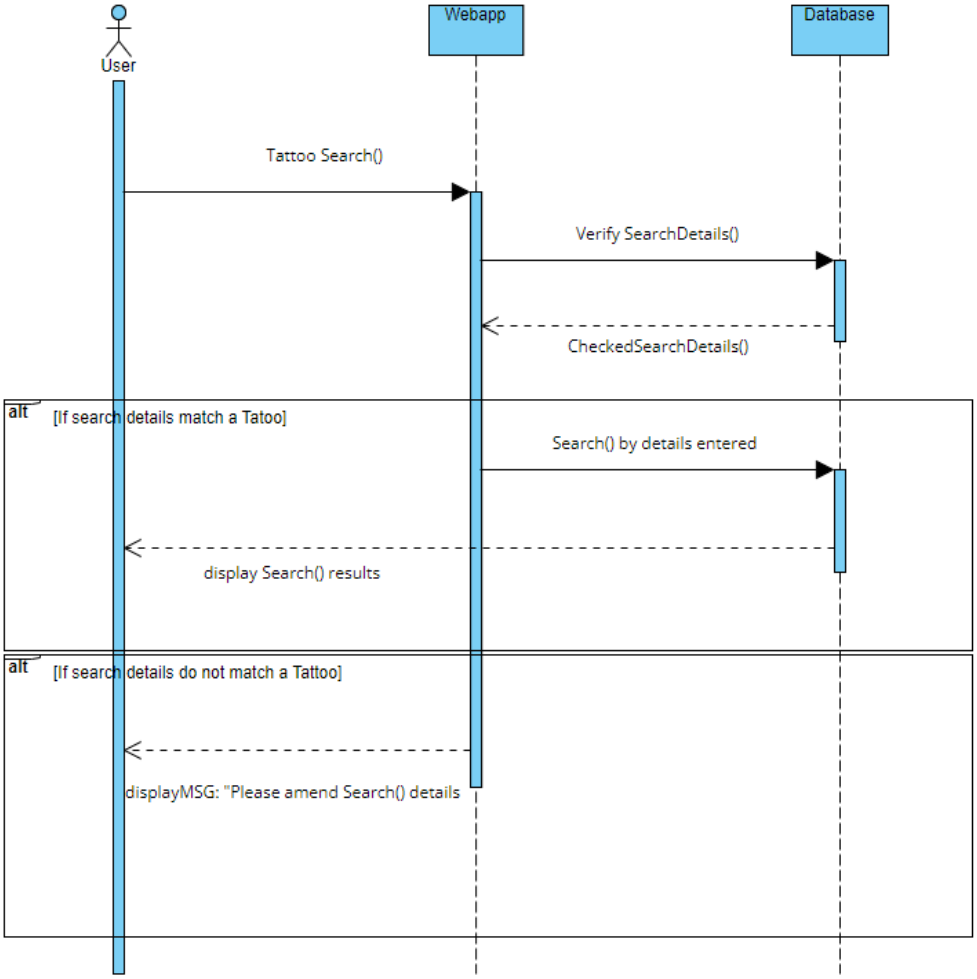

Figure 11 - Search function sequence diagram

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3.3.5. Figure 12 – Recommendation system sequence diagram

- This sequence diagram gives a visual representation of how the recommendation system will work.
- First the user will request to get a recommendation.
- After this the users historical tattoo ratings will be analysed, and the users' preferences will be computed based on similarities between tattoos, derived from a comparable list of saved and rated tattoos by users.
- From the users' preferences the recommendation is then calculated, and the most suitable recommendations will be displayed to the user.

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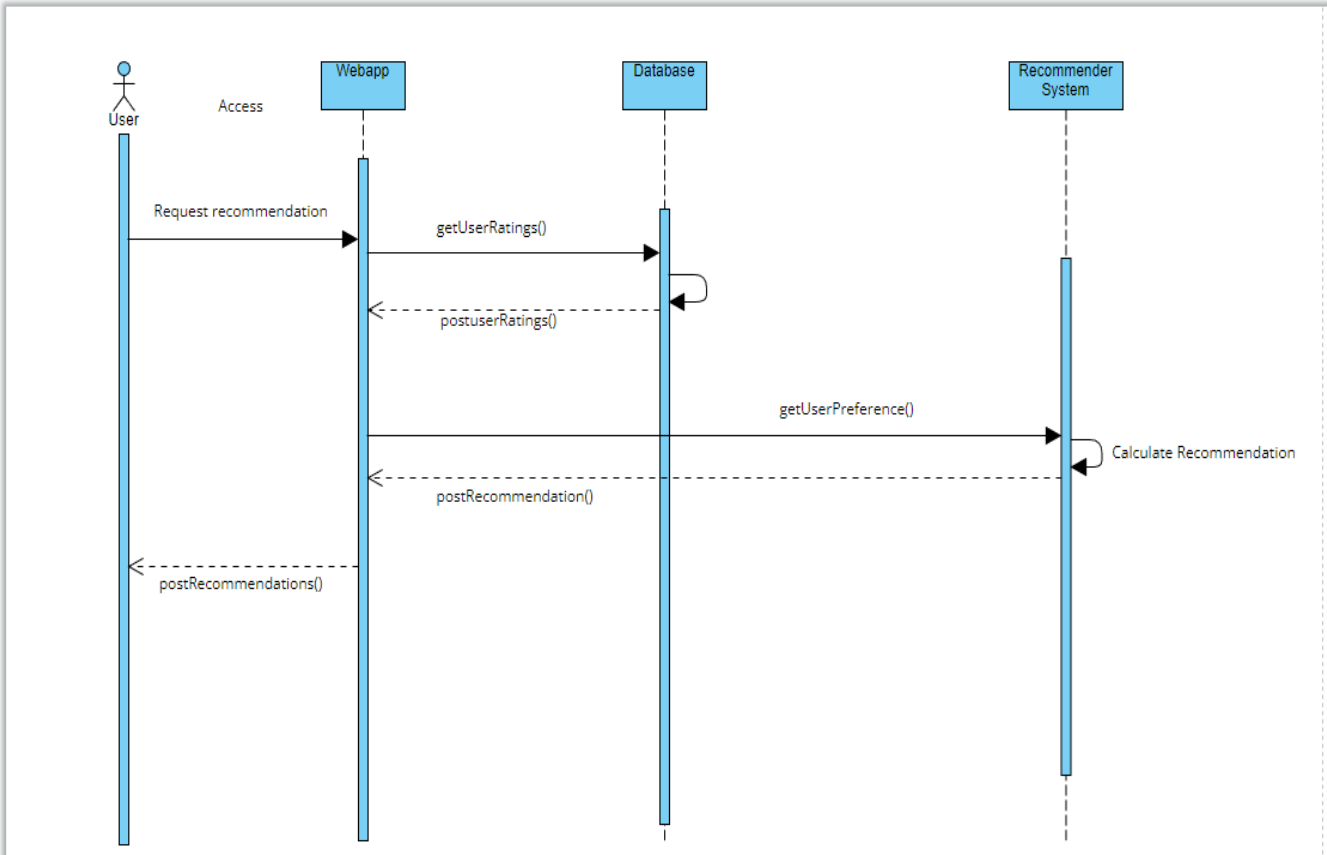

Figure 12 - Recommendation system sequence diagram

3.3.6. Figure 13 – Firebase Storage for Images

- Figure 13 represents how the data will be stored in image format for this project. Firebase Storage was utilized in this project for storing and managing tattoo images, due to its high-performance, secure, and scalable nature. Firebase Storage provides a simple and efficient way to upload, store, and retrieve images, making it an ideal solution for projects requiring multimedia handling. Additionally, it offers seamless integration with other Firebase services and the React front-end, streamlining the development process. By using Firebase Storage, the project ensured a reliable and efficient system for managing tattoo images, resulting in an enhanced user experience and easy access to visual content.

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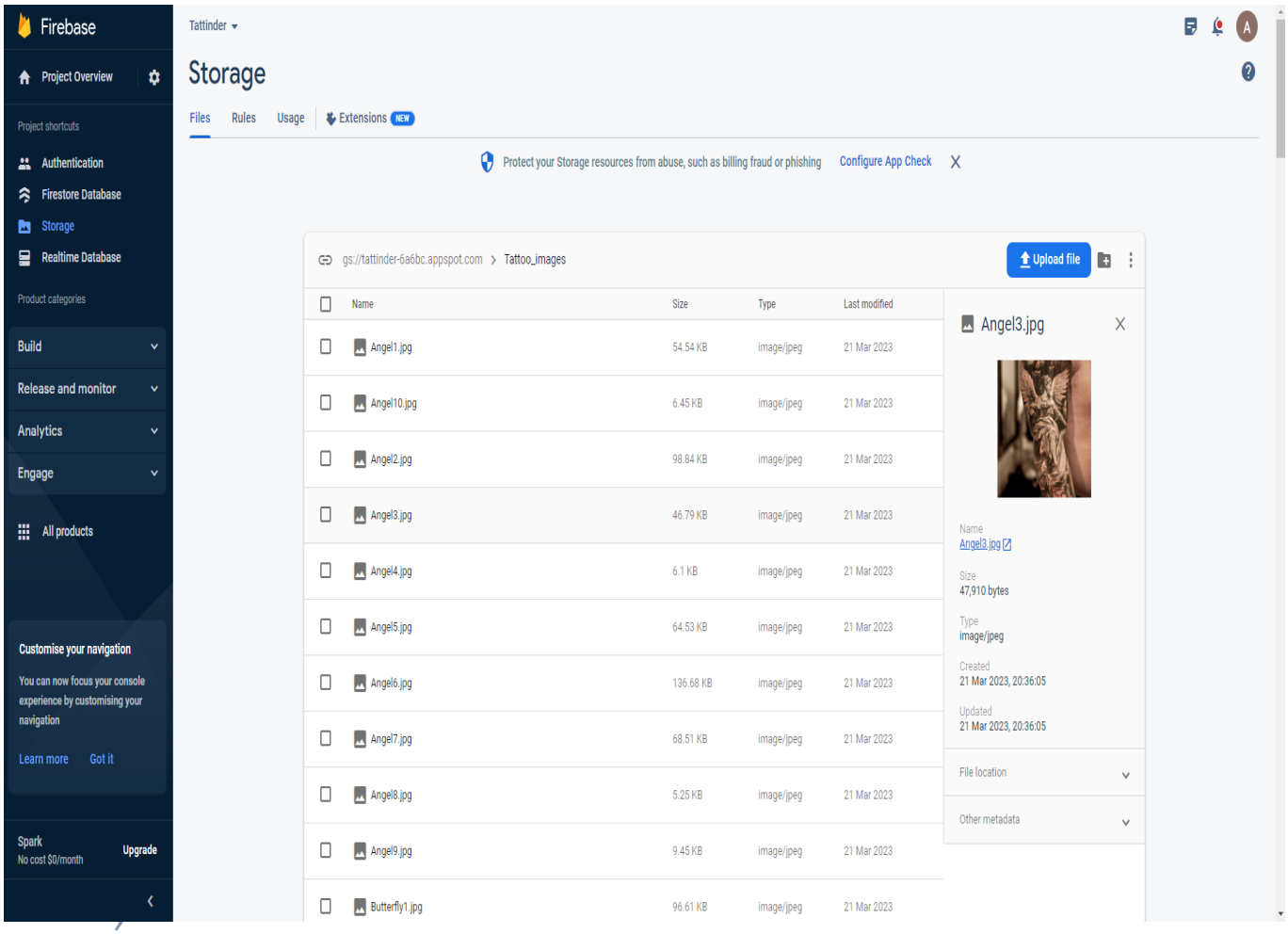

Figure 13 - Firebase Storage

3.3.7. Figure 14 – Firebase Realtime Database

Firebase Realtime Database was employed in this project for managing tattoo metadata, scalability, and ease of integration with the React front-end. The Realtime Database allows for efficient storage and retrieval of tattoo metadata, such as style, colour, and area of the body, in a well-structured JSON format. This real-time synchronization ensures that any updates to the metadata are instantly propagated to all connected clients, providing a seamless user experience. Furthermore, the JSON file structure is easily navigated and lightweight, allowing for efficient querying and manipulation of data. The JSON file was hard coded and imported into the Realtime database for the purposes of displaying tattoo information as well as serving as a content-based similarity metric by which the recommendation system would inevitably calculate. By utilizing Firebase Realtime Database for tattoo metadata management, the project achieved a dynamic, scalable, and responsive system for handling the essential information related to tattoos. The 'Tattoo_images' node in the below collection is the JSON file imported into firebase Realtime Database.

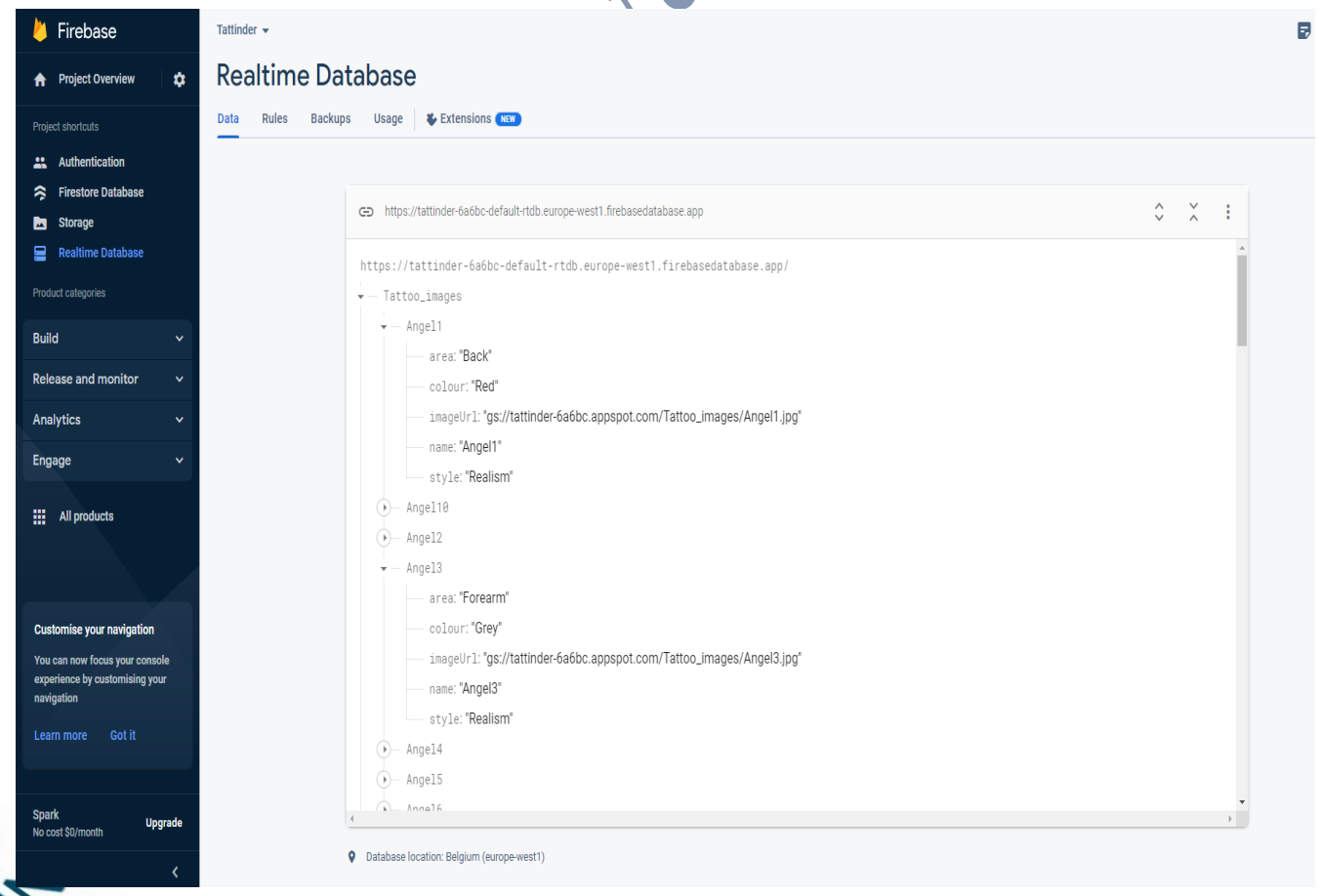

Figure 14 - Image Metadata

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3.3.8. Figure 15 – Firebase Realtime Database

In the project, a 'saved_tattoos' node was created within the firebase realtime database to store tattoo ratings and images for each user individually. This design choice allowed for efficient organization and management of user-specific data. By assigning a unique user ID to each user, the system ensured that their corresponding 'saved_tattoos' records were stored and retrieved effectively. When a user rates a tattoo, the system saves the rating along with the tattoo image and user ID under the 'saved_tattoos' node. This approach streamlines data management and enables personalized recommendations based on each user's preferences. Consequently, the 'saved_tattoos' node plays a critical role in enhancing the user experience by providing customized tattoo suggestions tailored to their unique tastes.

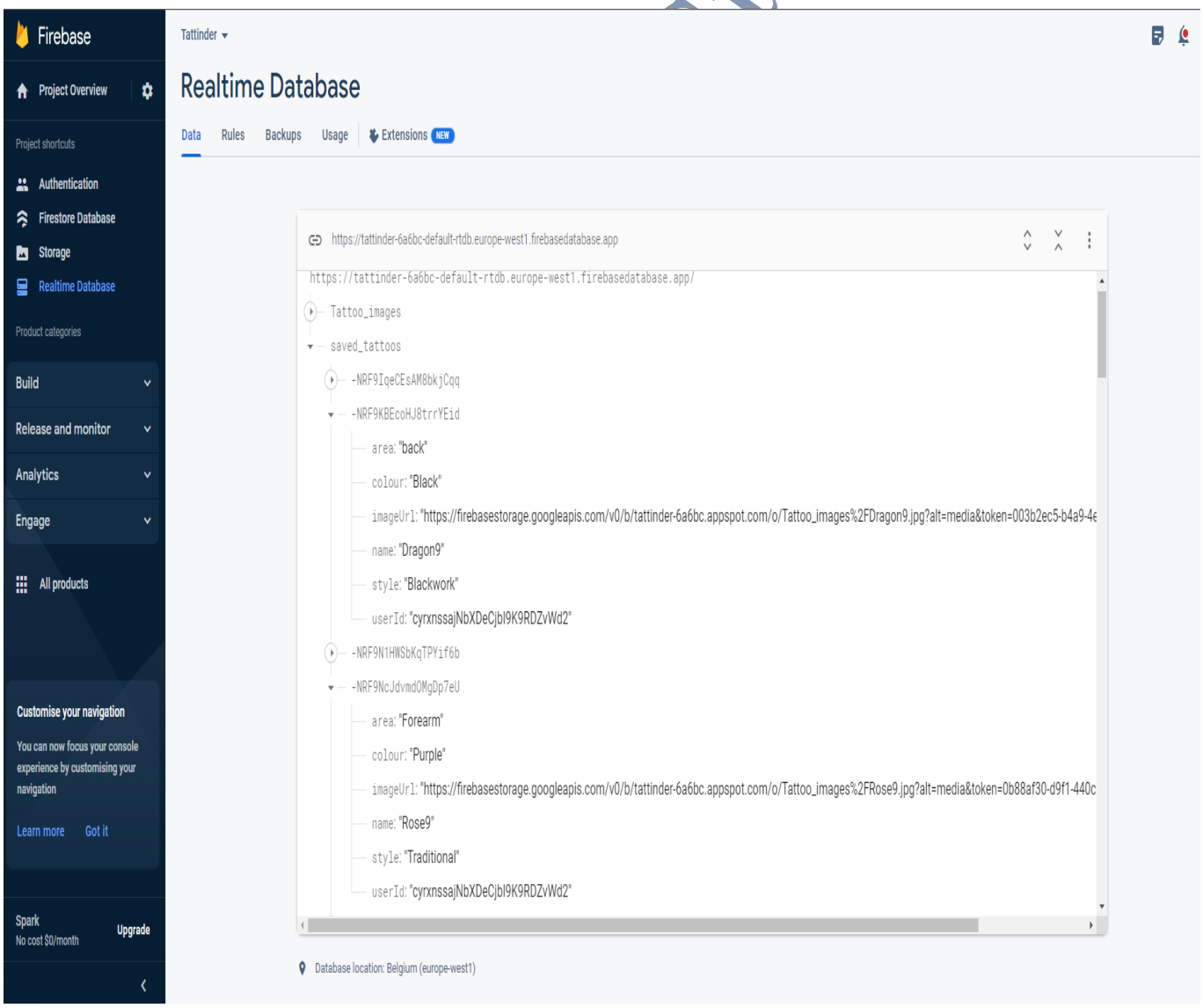

Figure 15 - User Tattoo Ratings

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3.3.9. Figure 16 – Firebase Authentication

In this project, Firebase Authentication was employed to manage user access, providing a secure and reliable solution for user identification and authorization. By incorporating Firebase Authentication, the project leveraged a comprehensive suite of authentication features, such as user registration, login, and password management, without the need for manual implementation. This not only saved time and effort during development but also ensured a robust security infrastructure. The use of Firebase Authentication allowed the application to seamlessly handle user accounts and data, while safeguarding sensitive information and maintaining a high level of security, ultimately enhancing the overall user experience and trust in the platform.

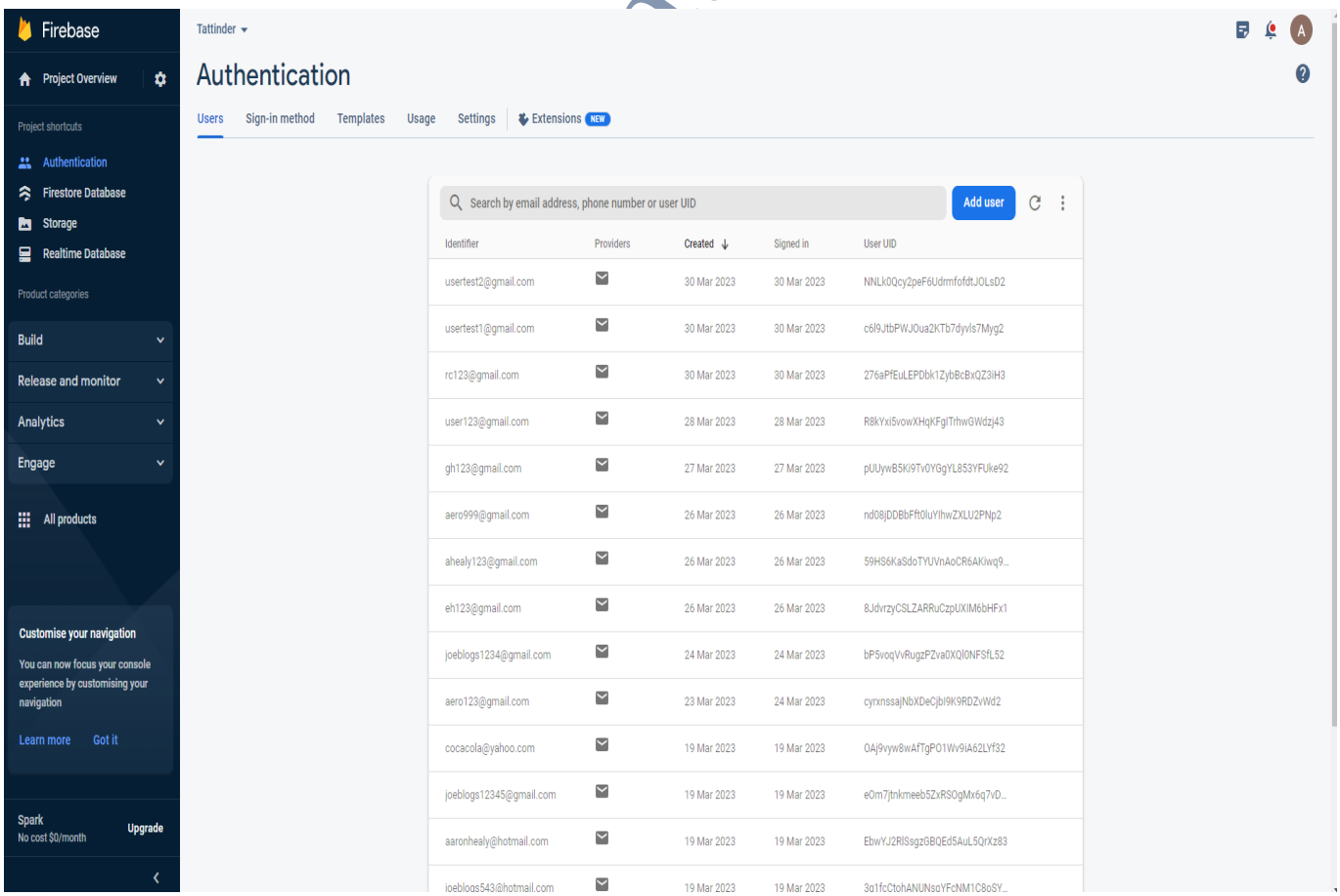

Figure 16 - Firebase User Authentication

Case Study Only

3.4. Webapp UI

3.4.1. Login Page:

The user interface (UI) of the login page was designed with simplicity and ease of use in mind, ensuring that users could quickly and effectively access the application. The clean, minimalistic layout featured clearly labelled input fields for email and password, as well as distinct buttons for signing in or navigating to different areas of the website, however error checking has been implemented so if a user is not signed in, they cannot navigate to any page except for login and register pages. Aesthetically pleasing and visually consistent with the rest of the application, the login page provided a welcoming and user-friendly environment, which encouraged users to engage with the application and facilitated a smooth onboarding experience.

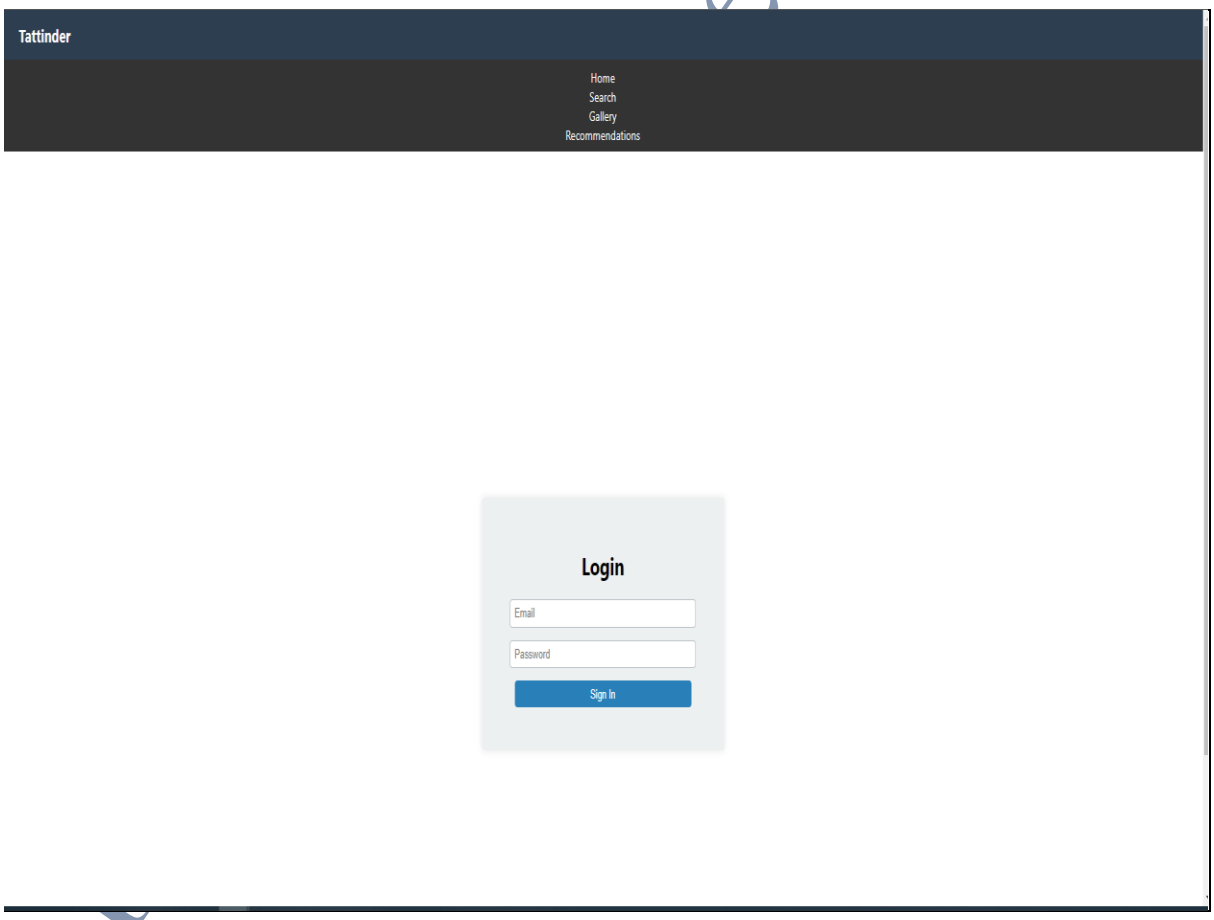

Figure 17 - Login Page

For Study Only

3.4.2. Register Page:

The user interface (UI) of the register page was designed to provide a seamless and straightforward experience for users looking to create a new account. The layout was kept clean and minimalistic, with clearly labelled input fields for essential information such as email, password, and the navigation bar. A prominent button for completing the registration process was also added. The overall aesthetic was visually consistent with the rest of the application, ensuring that users felt comfortable and confident while navigating through the registration process. By prioritizing usability and a cohesive design, the register page successfully facilitated user onboarding and encouraged engagement with the application.

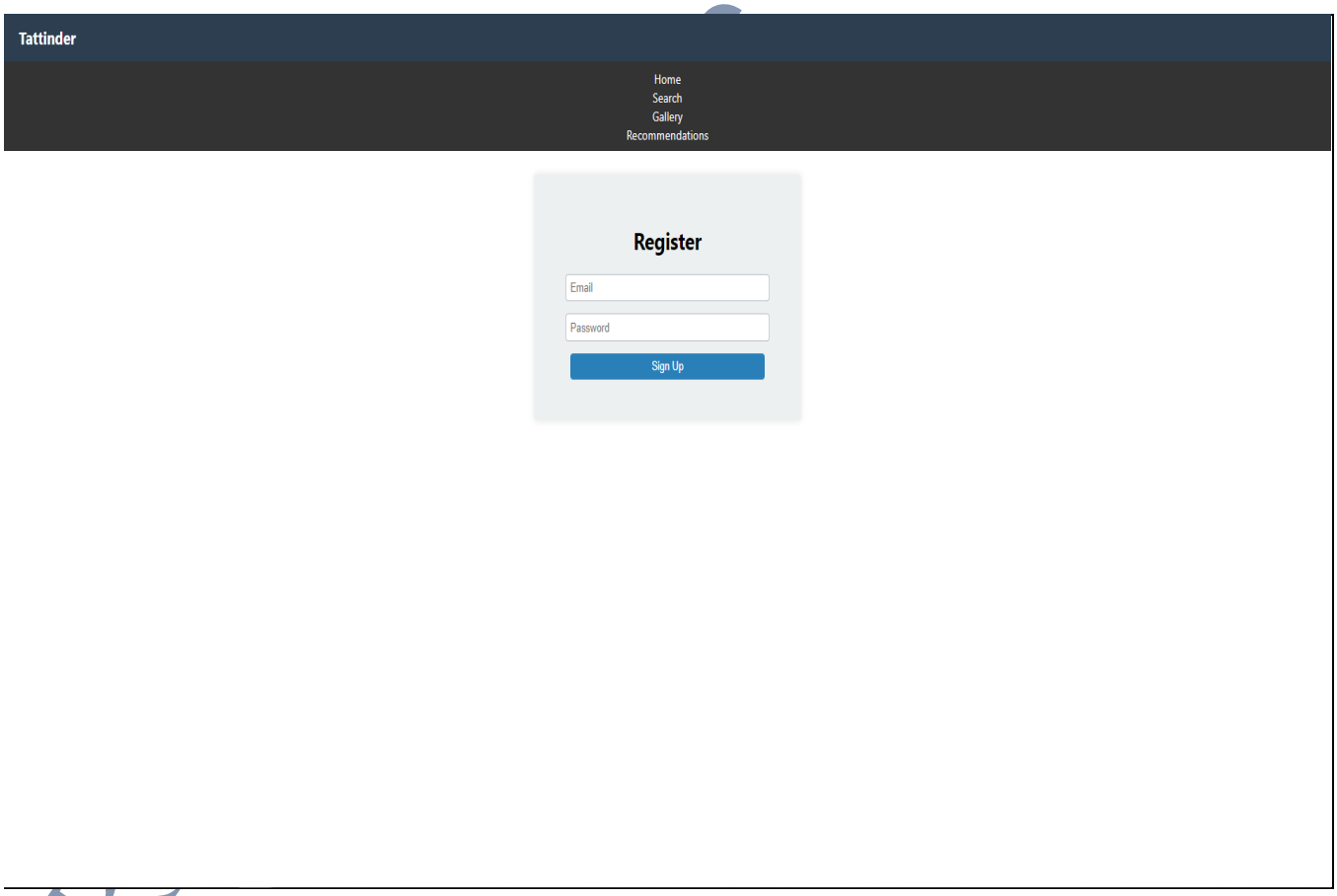

Figure 18 - Register Page

For Study Only

3.4.3. Home Page:

The user interface (UI) of the homepage was designed to create an inviting and engaging experience for users upon entering the application. The layout was thoughtfully organized, with an aesthetically pleasing and visually consistent design that seamlessly integrated with the rest of the application. Key features and navigation options were prominently displayed, making it easy for users to access essential functionalities such as search, gallery, and recommendations. The homepage also provided a brief overview of the application's purpose and its benefits to the users, encouraging them to explore further and engage with the platform. Overall, the homepage successfully served as a welcoming entry point that facilitated user navigation and interaction with the various features of the application.

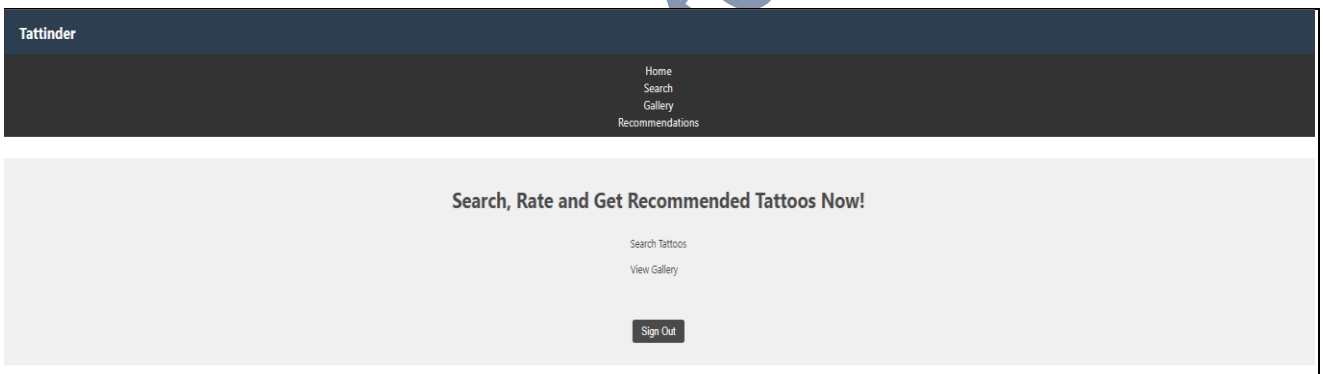

Figure 19 - Home Page

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3.4.4. Search & Rate Page:

The user interface (UI) of the Search Page was designed to offer users a convenient and efficient way to search for tattoos, providing an intuitive search experience. The page featured a search bar where users could easily input their desired search terms, along with a search button to initiate the process. The search results were presented in a visually appealing and organized manner, making it simple for users to browse through the displayed tattoos. Alongside each tattoo, users could find relevant information such as the tattoo's name, colour, and style. One of the key features of the Search Page was the integration of a rating system, which allowed users to rate tattoos based on their preferences. The rating system utilized star ratings, offering a familiar and user-friendly method for users to provide their feedback. Overall, the Search Page successfully combined an efficient search functionality with a visually pleasing layout, while enabling users to engage with the application by rating and thus saving their favourite tattoos.

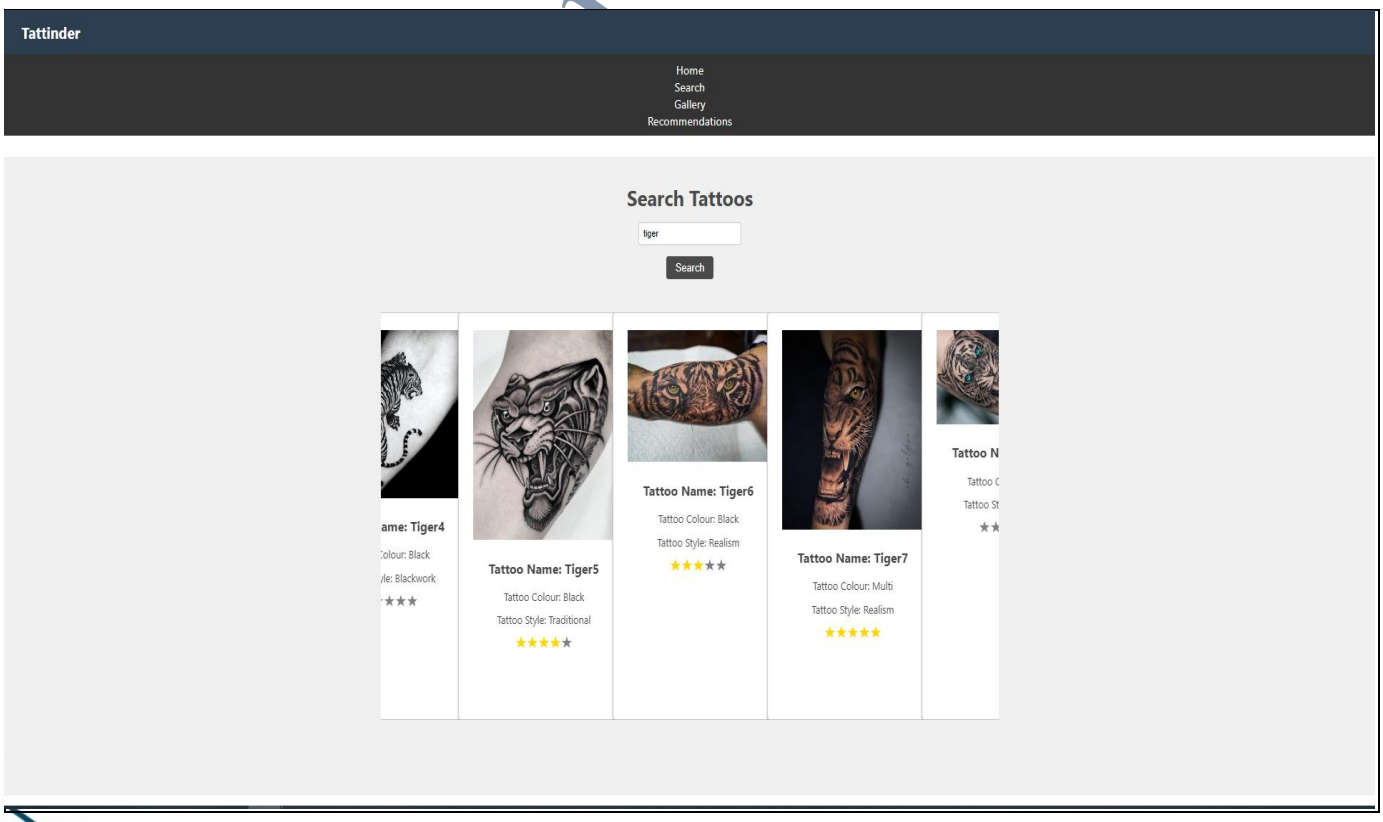

Figure 20 - Search & Rate Page

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3.4.5. Gallery Page:

The user interface (UI) of the Gallery Page was thoughtfully designed to showcase a visually appealing collection of tattoos, offering users an enjoyable browsing experience. The page displayed the tattoos in a well-organized grid layout, making it easy for users to navigate and explore various tattoo designs. Each tattoo was presented with a clear and high-quality image, ensuring that users could view the details of each design without difficulty. As users browsed the gallery, they could find relevant information about the tattoos, such as their names, colours, and styles. The Gallery Page aimed to inspire users by presenting a diverse range of tattoo designs and styles, encouraging them to explore and discover new ideas. Overall, the Gallery Page successfully combined an aesthetically pleasing presentation with a user-friendly layout, providing a seamless and engaging browsing experience for users, allowing them to view previously rated tattoos effortlessly.

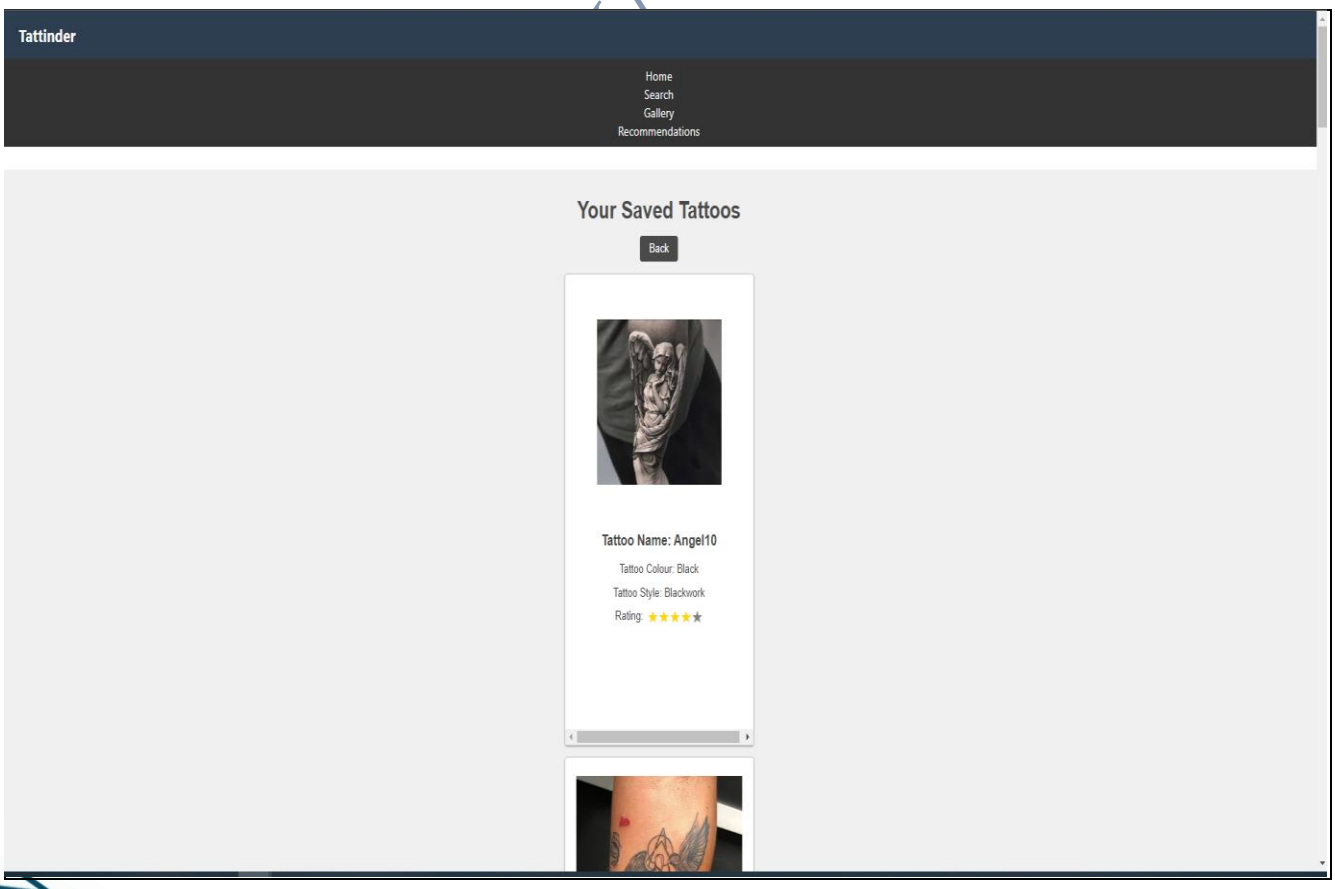

Figure 21 - Gallery Page

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3.4.6. Recommendations Page:

The user interface (UI) of the Recommendations Page was carefully designed to provide personalized tattoo recommendations for each user, based on their unique preferences and interests. This page displayed a list of tattoos tailored to the individual user, taking into account their past ratings and preferences for styles, colours, and areas. The recommendations were presented in an easy-to-navigate layout, allowing users to seamlessly browse and explore suggested tattoo designs. Each recommended tattoo was accompanied by a clear and high-quality image, as well as relevant information about the design, such as its name, colour, and style. Additionally, users could view the average ratings for each tattoo, which were calculated based on the preferences of other users with similar interests. The Recommendations Page aimed to offer a personalized and engaging experience, helping users discover new tattoo designs that aligned with their tastes. In summary, the Recommendations Page successfully combined a user-friendly interface with a content-based recommendation system, providing a valuable and enjoyable browsing experience for users.

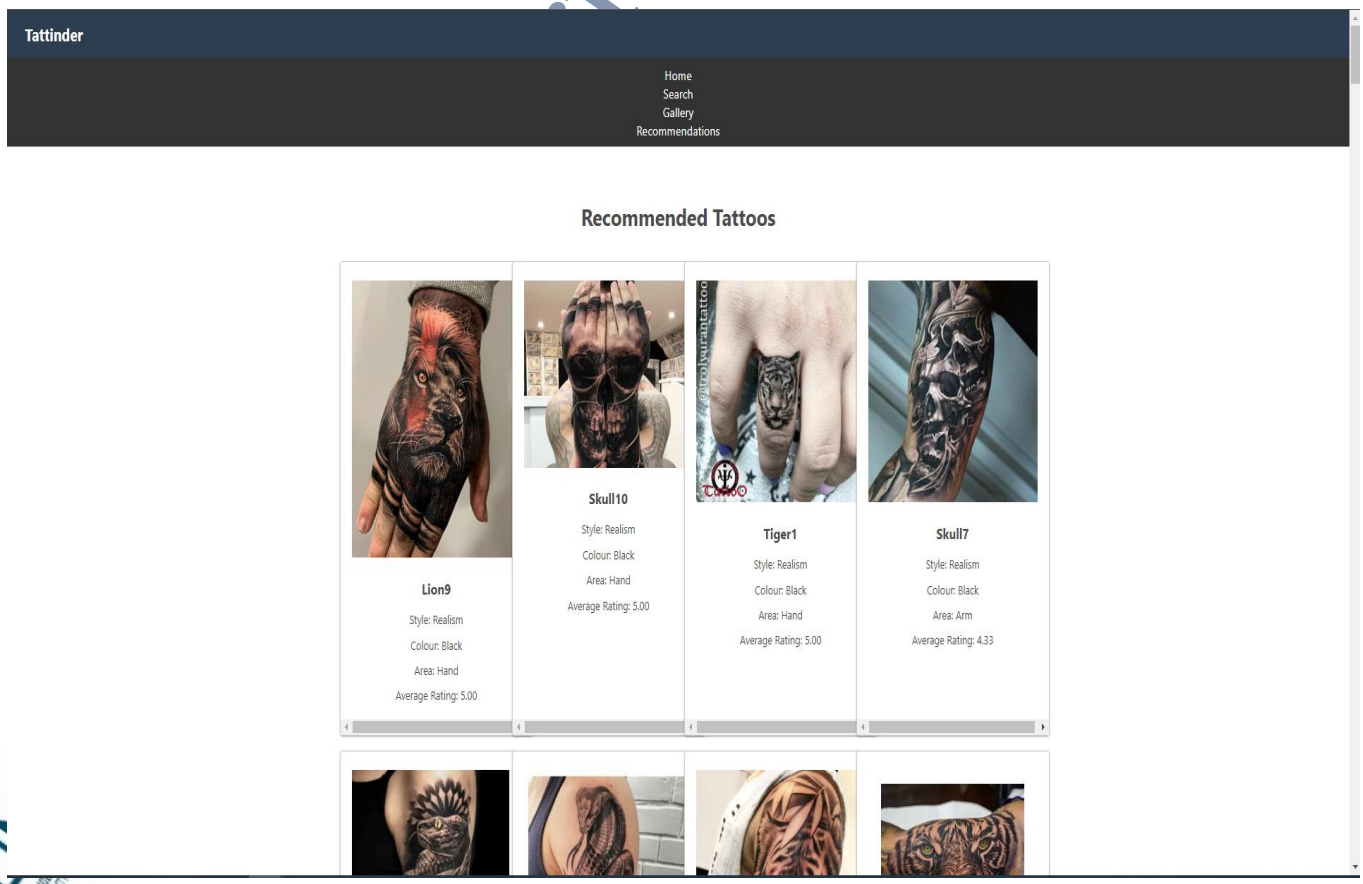

Figure 22 - Recommendations Page

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3.5. Conclusions

In conclusion a three-tier architectural system using React for front-end & back-end logic, along with firebase for the database tier will be the overall design for this project. Three-tier architectural system are often used in projects of this size and scale. React and Firebase were chosen for this project due to their efficiency, scalability, and ease of use. React, a popular JavaScript library, offers a component-based approach to building user interfaces, making it easier to manage and maintain the code. It also provides excellent performance and a rich ecosystem of libraries and tools, which facilitate the development process.

Firebase, on the other hand, is a robust backend-as-a-service platform that simplifies the process of setting up and managing databases and authentication systems. It offers real-time syncing capabilities, making it an ideal choice for projects requiring responsive and dynamic user experiences. Additionally, Firebase's seamless integration with React allows for a smooth development process, further justifying their selection for this project.

Together, React and Firebase provided a powerful and efficient combination of new technologies that enabled the successful implementation of the desired features while maintaining a high level of performance and user experience.

An example of the three-tier architectural system can be seen in figure 23 (21) below:

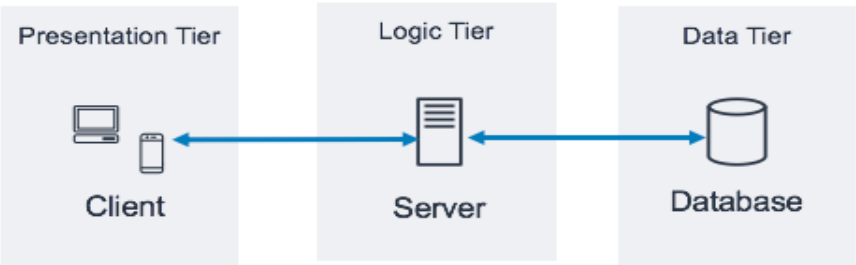

Figure 23 - Three-tier architectural system (21)

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4. Experimental Development

4.1. Introduction

In this experimental development project, the aim was to create a web application for tattoo enthusiasts that allows them to save their favourite tattoo images and receive personalized recommendations based on their preferences. Throughout the development process, various challenges were encountered related to the implementation of the user interface, back-end storage, and recommendation algorithm. The primary focus of the experimentation is to find the most effective solutions to these challenges while maintaining a user-friendly and efficient application. By carefully analysing the requirements and iteratively refining the code, the project strives to deliver a robust web application that provides users with a seamless browsing experience and tailored tattoo recommendations.

4.2. Software Development

The development of the tattoo web application involved several stages, including front-end design, back-end implementation, recommendation algorithm development, and integration of all components. In this section, these stages will be discussed in detail, focusing on the challenges encountered and the solutions implemented.

4.3. Front-End Implementation

The front-end design of the web application plays a crucial role in providing a user-friendly experience. The primary goal was to create an intuitive interface that allows users to easily browse, save, and rate tattoo images. To achieve this, React.js, a popular front-end library for building user interfaces was used.

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React.js enables the creation of reusable components, allowing for efficient code organization and easier maintenance. The application's main components include the login/register page, the browsing page, the gallery page, and the recommendations page.

The login/register page uses Firebase Authentication to handle user authentication securely. Users can register for an account and subsequently log in. The browsing page displays tattoo images fetched from the Firebase Realtime Database and allows users to rate and save their favourites. The gallery page allows users to view their previously saved tattoos along with the tattoos information and rating the user gave the tattoo. The recommendations page uses a content-based recommendation algorithm to provide personalized tattoo suggestions based on the user's saved and rated tattoos.

The front-end design involved creating responsive and visually appealing components while maintaining a consistent look and feel throughout the application. CSS was used to style the components, ensuring a seamless and responsive design across different screen sizes and devices.

4.4. Middle-tier Implementation

The middle-tier architecture of the tattoo recommendation web application serves as a bridge between the front-end user interface and the back-end database system. This layer is responsible for handling logic, processing user input, managing data flow, and communicating with external services, such as Firebase Realtime Database and Firebase Storage. In this project, the middle-tier is primarily composed of React components and functions that interact with Firebase services to provide a seamless user experience.

The middle-tier architecture can be broken down into the following key components:

React Components: These are the building blocks of the application, representing various parts of the user interface. The components include the Login and Register pages for user

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authentication, the Search Page for browsing and searching tattoos, the gallery page for viewing previously rated and saved tattoos, the Recommendations Page for generating personalized tattoo recommendations, and other auxiliary components for navigation and layout.

State Management: The application uses Reacts built-in state management system (useState hook) to manage the application state, such as user authentication status, search results, and recommendations. This allows the components to respond to changes in the data and update the user interface accordingly.

Asynchronous Data Fetching: The application uses asynchronous functions to fetch data from the Firebase Realtime Database and Firebase Storage. These functions use the Firebase SDK's APIs to retrieve data, such as tattoo images and user ratings, and update the application state. They also handle error scenarios, ensuring that the application remains stable even when faced with unexpected issues.

Data Processing and Recommendation Algorithm: The middle-tier architecture includes functions responsible for processing the data retrieved from the back-end, such as filtering search results and calculating average ratings for recommendations. The recommendation algorithm is content-based, taking into account the user's ratings and tattoo attributes, such as style, colour, and body area, to generate personalized recommendations.

Data Persistence: The middle-tier architecture interacts with the Firebase Realtime Database to store and retrieve user ratings, saved tattoos, and user information. This allows the application to maintain a persistent state across sessions, enabling users to access their data and recommendations from any device.

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In summary, the middle-tier architecture of the tattoo recommendation web application is responsible for orchestrating the flow of data between the front-end user interface and the back-end data storage system. By leveraging modern web technologies, such as React and Firebase, the middle-tier ensures a smooth and responsive user experience, while also handling the complex logic and data processing required to deliver accurate and personalized tattoo recommendations.

4.5. Back-End Implementation

The back-end of the tattoo web application is responsible for storing and managing tattoo images, user accounts, and user preferences. Firebase was chosen, a serverless platform by Google, for its ease of use, scalability, and real-time data synchronization capabilities.

Firebase Realtime Database was employed to store tattoo images and user preferences. Firebase Authentication managed user accounts and authentication, ensuring the secure handling of user data. Firebase Storage provided a storage solution for tattoo images, allowing easy retrieval of image URLs for display on the front-end.

Integrating Firebase with the React.js front-end involved the use of several Firebase SDK APIs, such as 'firebase/auth', 'firebase/database', and 'firebase/storage'. These SDK APIs facilitated communication between the front-end and back-end, enabling real-time data fetching, storage, and updates.

To fetch tattoo images and their associated data, Firebase Realtime Database references were created using the 'ref' function. The 'onValue' function was then employed to listen for data changes in real-time, ensuring that the front-end always displayed the most up-to-date information.

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In addition, Firebase Storage was used to handle image storage and retrieval. This involved converting Google Storage (gs://) URLs to HTTPS URLs, which could be easily displayed on the front-end. A custom utility function, 'getImageUrl', was created to handle this conversion.

4.6. Recommendation Algorithm Development:

The core complexity of the tattoo web application lies in its ability to provide personalized tattoo recommendations based on user preferences. To achieve this, a content-based recommendation algorithm was developed.

The content-based recommendation algorithm operates by calculating average ratings for each tattoo in the database, considering only the ratings from users who have similar preferences to the current user. Similarity between users is determined based on matching tattoo style, colour, and area attributes. The algorithm then sorts the tattoos by their average rating and returns the top recommendations to the user.

The algorithm was implemented as a function, 'calculateAverageRatings', which takes in two arguments: ' allTattoos' and 'userRatings'. 'allTattoos' represents a list of all available tattoos in the database, while 'userRatings' contains the ratings provided by the current user for their saved tattoos.

The 'calculateAverageRatings' function first maps over the 'allTattoos' array, comparing each tattoo to the 'userRatings' data. It filters out the user ratings with similar tattoo attributes (style, colour, and area) using the 'filter' function. This results in a list of similar tattoos rated by the current user.

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Next, the function calculates the sum of the ratings of these similar tattoos using the 'reduce' function. The average rating is then computed by dividing the sum by the number of similar tattoos. If there are no similar tattoos, an average rating of zero is assigned.

The function returns a new array of tattoo objects, each containing the original tattoo data along with the calculated average rating. Finally, this new array is sorted in descending order based on the average rating, resulting in a list of top-recommended tattoos for the current user.

In summary an explanation of the algorithm is as follows:

1. Fetch all tattoo data and the user's saved tattoos (with ratings).
2. Calculate average ratings for each tattoo considering only similar tattoos (with matching style, colour, and area) rated by the user.
3. Sort the tattoos by their calculated average ratings.
4. Display the sorted tattoos as recommendations.

4.7. Project Challenges

Throughout the implementation of the project, several challenges were encountered that required careful consideration and creative problem-solving. This document will outline some of the most significant challenges faced during the development process.

Data collection and storage: One of the primary challenges in building the webapp was gathering a diverse and comprehensive collection of tattoo images for users to browse and like. It was essential to ensure that the database had a wide variety of tattoo styles, colours, and body areas to cater to different user preferences. Additionally, it was necessary to devise an efficient and scalable storage solution for these images, as well as for user data, such as liked tattoos and recommendations.

User authentication and security: As the webapp required users to create accounts and login to access their personalized recommendations, implementing a secure and robust user

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authentication system was crucial. Ensuring that user data remained confidential and protected from unauthorized access posed a significant challenge, requiring the integration of a reliable authentication system, such as Firebase Authentication, and adherence to best practices for secure data storage.

Image loading and optimization: Due to the large number of high-quality tattoo images stored in the database, optimizing image loading and display for the webapp was a considerable challenge. Loading all images at once could lead to slow page load times and a poor user experience. To overcome this, image optimization methods were used, such as resizing and compression, to minimize the impact of image loading on the webapp's performance.

Building a recommendation system: Designing and implementing a recommendation system that could accurately provide personalized tattoo suggestions based on users' preferences was a complex task. The system needed to analyse user ratings for various tattoo attributes, such as style, colour, and body area, and generate recommendations by identifying patterns and similarities among liked tattoos. This required the development of an algorithm that could calculate average ratings for each tattoo and sort them based on their relevance to the user's preferences.

Responsiveness and cross-platform compatibility: Ensuring that the webapp functioned smoothly across various devices and platforms was another significant challenge. It was necessary to design and develop a responsive user interface that could adapt to different screen sizes and resolutions, providing an optimal browsing experience for all users, regardless of their device.

Testing and evaluation: Thorough testing and evaluation of the webapp's functionality, performance, and user experience were critical to identifying and resolving any issues or shortcomings. This involved conducting incremental testing throughout the development process, as well as gathering user feedback to improve the webapp's design and functionality.

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Time management and prioritization: Due to the numerous components and features involved in building the webapp, managing time effectively and prioritizing tasks were essential to ensuring the project's success. Balancing the development of the front-end user interface, back-end logic, and database management required careful planning and organization.

Overall, the development of the project presented several challenges that required careful consideration, planning, and problem-solving. Overcoming these obstacles involved the utilization of various technologies and techniques, such as image optimization, secure authentication systems, and algorithm development, to create a functional and user friendly webapp for browsing and recommending tattoos.

4.8. Conclusions

In conclusion, the software development process for the tattoo recommendation web application has been a challenging yet rewarding experience. Throughout the project, various design principles, software development methodologies, and technologies have been employed to create a user-friendly and effective platform for browsing tattoos and generating personalized recommendations. The project has successfully implemented an array of functionalities, such as user authentication, image storage and retrieval, search features, and a content-based recommendation algorithm.

The development process was iterative, with each phase building upon the previous one to refine the application and address any issues encountered. By following best practices and using well-established software development methodologies, such as Agile, the project developed into a robust and scalable application that can be easily maintained and improved over time. The application leverages modern web technologies, such as React,

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Firebase, and the Firebase Realtime Database, to deliver a seamless and responsive user experience.

Throughout the development process, the focus was on ensuring that the application met the users' needs, with a particular emphasis on usability, performance, and the accuracy of the recommendation algorithm. By thoroughly testing and evaluating the system, it was simplistic to identify areas for improvement and make necessary adjustments, ultimately delivering a high-quality product.

The completion of the software development phase marks a significant milestone in the project. However, it is essential to recognize that software development is an ongoing process, with the need for continuous refinement, updates, and enhancements. As the user base grows and new technologies emerge, the application will need to evolve and adapt to meet the changing needs of its users.

In summary, the software development process for the tattoo recommendation web application has been a journey of exploration, learning, and growth. Through diligent planning, execution, and evaluation, a platform has been created that effectively addresses the project's aims and objectives, providing users with a valuable tool for discovering tattoos and receiving personalized recommendations based on their unique preferences.

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5. Testing and Evaluation

5.1. Introduction

In this chapter, the testing and evaluation methodologies applied to the tattoo recommendation web application project will be discussed. The primary goal of the testing and evaluation phase is to ensure that the developed system meets the project aims and objectives, effectively providing a user-friendly platform for browsing tattoos and generating personalized recommendations based on users' preferences. To achieve this, it is essential to thoroughly evaluate the web application's performance, user experience, and recommendation accuracy.

Testing and evaluation are integral components of the software development life cycle, contributing to the overall quality and reliability of the final product. For this project, a combination of manual and automated testing techniques will be employed to assess the system's functionality, usability, and accuracy of recommendations. This multi-faceted approach will help identify potential issues, make necessary improvements, and validate the system's effectiveness in meeting user needs.

During the testing phase, various aspects of the web application will be evaluated, including the user interface (UI), back-end implementation, and the recommendation algorithm's performance. The testing process will involve a series of functional, integration, and system tests to ensure that all components are working correctly and that the application's performance aligns with the initial project requirements. Additionally, the testing phase will include usability tests to assess the user experience and gather feedback for potential improvements.

To evaluate the accuracy of the recommendation system, qualitative methods were employed. Qualitative evaluation will involve gathering user feedback through surveys and interviews, assessing the overall satisfaction with the recommendations and the perceived usefulness of the system.

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Throughout the testing and evaluation process, iterative improvements will be made to the web application based on the feedback and findings. This will ensure that the final product is not only functional and user-friendly but also capable of providing accurate and personalized tattoo recommendations. By closely monitoring the system's performance and user feedback, continuous refinements could be made to the web application all while striving to achieve the project's overall objective: delivering a simplistic, user-friendly platform for browsing tattoos and generating meaningful recommendations.

5.2. Testing and Issues

Task Name	Task Description	Start Date	End Date	Estimated time to finish	Passed/Fail
Connect Database layer	Connect database to webapp and perform crud operations	5th January2023	January 9th 2023	6 hours per day for 4 days	Passed
Login/Register	Test the login and register functionality	16th January 2023	18th January 2023	A couple of hours on any day from 16th -17th -18th	Passed
GUI	Test the interface to ensure quality of life for users	16th January 2023	18th January 2023	A couple of hours on any day from 16th -17th -18th	Passed
Search Function	Ensure the search function is fetching results as desired.	25th January 2023	26th January 2023	5-10 minutes on the 25th to test.	Failed 26th January
Save Function	Test save tattoo function is working as intended	10th February 2023	13th February 2023	5-10 minutes on the 10th to test saved tattoos.	Failed 13th January
Recommender System	Test recommendations on different users to measure the outputs.	1st March 2023	3rd March 2023	2-3 Hours each day to ensure cold start problems are not present.	Failed 3rd March

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5.2.1. Search Page Testing Issues:

One specific reason for the initial failure of the Search Page implementation was the difficulty in fetching the tattoo images from Firebase Storage and displaying them correctly on the web page. The challenge arose from the fact that Firebase Storage uses a different method to access stored files, requiring the generation of a download URL to fetch and display images.

Initially, the page attempted to directly use the image references stored in Firebase Realtime Database, which would not have worked as expected because these references are not directly usable as image URLs. To address this issue, the Firebase Storage API was integrated and used the getDownloadURL() function, which takes the storage reference for a specific image and returns a URL that can be used to fetch the image.

The solution involved creating a separate function, fetchImageURL(gsUrl), which accepts the storage reference (gsUrl) for an image, retrieves the corresponding download URL using the getDownloadURL() function, and returns the download URL. This function was then called for each tattoo in the search results, updating the imageUrl property of the tattoo object with the download URL before displaying the images on the Search Page. By correctly integrating the Firebase Storage API and handling the asynchronous nature of the image retrieval process, the page could then fetch and display the tattoo images as intended, ultimately resolving the issue, and making the Search Page functional.

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5.2.2. Save and Rate Issues:

When the save and rate functions were first implemented for the tattoo webapp, a few issues related to the structure of the data and the way the Firebase Realtime Database was interacted with were encountered. Here's an explanation of what went wrong and the steps that were taken to resolve the issues:

Data structure and organization: Initially, the tattoo ratings were stored as part of the tattoo object itself, which caused issues when multiple users rated the same tattoo. This resulted in the last rating overwriting any previous ratings, making it impossible to calculate accurate average ratings for recommendations.

Solution: To solve this issue, a separate 'saved_tattoos' node in the Realtime Database was created to store the user-specific tattoo ratings. By doing so, each user's rating could be stored independently, allowing for the calculation of average ratings based on multiple users' preferences.

Saving and rating functionality: When the save and rate functions were first implemented, separate functions were used to handle saving and rating tattoos. This led to confusion and potential errors in managing the data and updating the database.

Solution: The saving and rating functionalities were combined into a single function, handleRatingChange(tattoo, newRating), which streamlined the process of updating the tattoo ratings. This function accepts the tattoo object and the new rating as arguments, then saves the rating information (along with the user ID) to the 'saved_tattoos' node in the Realtime Database.

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5.2.3. Recommender system Issues:

During the implementation of the recommendation system, an issue arose related to the calculation of average ratings for tattoos. The function calculateAverageRatings was designed to calculate the average rating of tattoos based on the user's ratings of tattoos with similar attributes, such as style, color, and body area. However, there were concerns regarding the appearance of NaN (Not a Number) values in the recommendations.

Upon further examination, it was discovered that the NaN values occurred when there were no similar tattoos found in the user's rated tattoos, and an attempt was made to calculate the average rating of an empty set. To address this issue, a conditional statement was added to check if the length of the similar tattoos array was greater than zero before attempting to calculate the average rating. This change ensured that average ratings were calculated only when there were valid user-rated tattoos with similar attributes, preventing NaN values from appearing in the recommendations.

The recommendation system in the code essentially utilizes a content-based filtering approach, as it considers the user's ratings of tattoos with similar attributes. With the fix applied, the recommender system now functions as intended, providing users with relevant tattoo recommendations based on their preferences.

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5.3. Evaluation

5.3.1. Smoke Testing:

Smoke testing (22) is a crucial stage of testing for any project. It falls under the umbrella category of system tests and lets the developer test for any critical flaws which may cause the system to crash. By using this test method, the developer can ensure all functionality is working correctly before proceeding to finely test each individual piece of software.

5.3.2. Useability Testing:

After the smoke test passed a usability test(23) was employed. In this test the developer registered to the webapp as if they were a first-time user. At this point the developer started to use the interface of the web app to try get a feel for how the web app functions in terms of quality of life. By using this test valuable insights were gained into the possible UI changes that were implemented along with quality of life changes for the user.

5.3.3. White Box Testing:

On top of these two testing methods, white box testing (24) was also used throughout the development cycle. White box testing refers to the method of testing an application while knowing all the inner code and functionalities. By using this method, the recommendation system was tested vigorously throughout development to ensure the most optimal similarity were chosen when making a recommendation.

5.3.4. Integration and testing:

Integration involved combining the various components and functionalities of the application, including front-end design, back-end implementation, and recommendation algorithm. This required ensuring seamless communication between the components and efficient data handling.

The integration process involved updating the 'RecommendationsPage' component to fetch tattoos and ratings, calculate recommendations using the custom recommendation

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algorithm, and display the recommended tattoos to the user. This was achieved by creating a 'fetchTattoosAndRatings' function that takes the user ID as an argument and updates the application state with fetched tattoos and recommendations.

During the development process, incremental testing was performed to ensure the proper functioning of each component and feature. This allowed for early identification of issues and the implementation of necessary fixes.

5.3.5. Some of the challenges encountered during the testing phase included:

Ensuring real-time data synchronization: One challenge was ensuring that the application always displayed the most up-to-date information from the Firebase Realtime Database. This was resolved by using the 'onValue' function to listen for data changes and update the application state accordingly.

Handling null values and exceptions: The application needed to handle cases where data was unavailable or incomplete, such as when a user has not yet saved or rated any tattoos. To address this, conditional statements and error handling were added to the code to manage null values and prevent potential crashes.

Authenticating users securely: Ensuring that user data was securely managed and protected required implementing proper authentication methods. This was achieved through the use of Firebase Authentication, which provides a secure and scalable solution for user management.

Image URL conversion: Retrieving image URLs from Firebase Storage involved converting Google Storage (gs://) URLs to HTTPS URLs. A custom utility function, 'getImageUrl', was

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created to handle this conversion, allowing for the seamless display of images on the front-end, this is because by default Google Storage is private per user, meaning it was necessary to convert the images to https URLs to ensure they were displayed as desired.

Optimizing performance: As the application grows in size and complexity, it is essential to maintain optimal performance. This involved careful consideration of the application's architecture, data handling methods, and component rendering to minimize any potential bottlenecks or performance issues.

Once the integration and testing phase was completed, the final tattoo web application provided users with an intuitive and user-friendly platform to browse, save, and rate tattoos, as well as receive personalized recommendations based on their preferences.

5.4. Google Lighthouse testing

Google Lighthouse was utilized as a key performance analysis tool throughout the development of this project. As an open-source, automated tool designed for improving the quality of web pages, Lighthouse provided valuable insights into the application's performance, accessibility, search engine optimization (SEO), and best practices. By regularly running Lighthouse tests during the development process, areas of improvement were identified, and subsequent optimizations were made, ensuring that the final application met high standards of performance and user experience. The consistent use of Lighthouse testing allowed for potential issues to be solved early on and continuous refinements to the application to be made, in order to achieve an optimal balance between functionality, performance, and user satisfaction.

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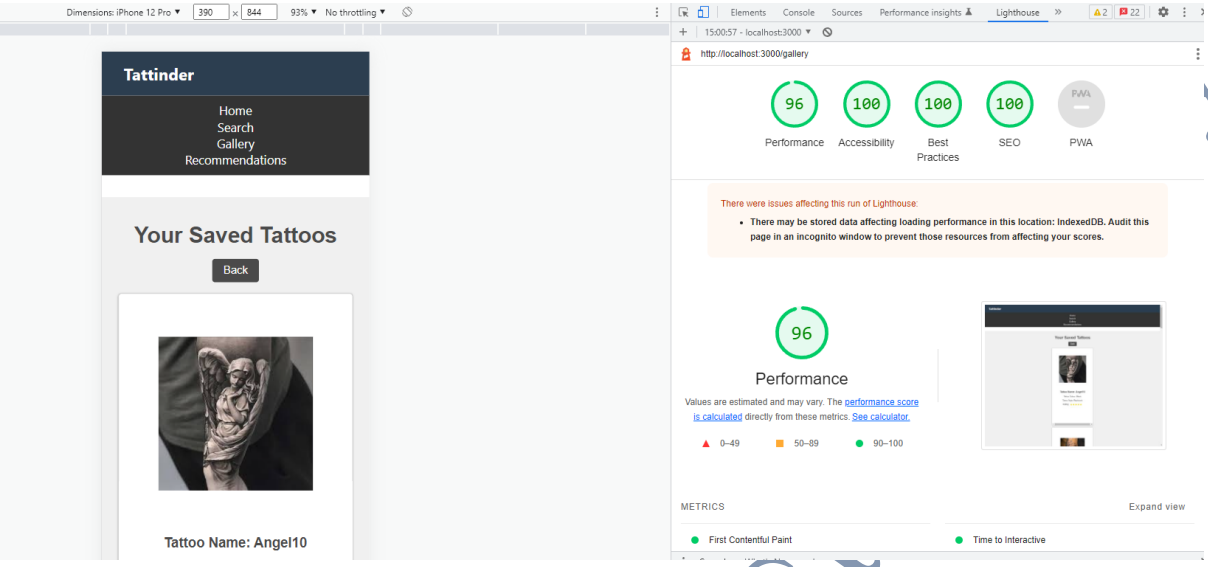

Figure 24 - Google Lighthouse

5.5. Conclusions

In conclusion, incremental testing and careful attention to detail ensured that the application was free of bugs and performance issues. The final product offers users an innovative and engaging way to explore and save tattoo images and receive recommendations tailored to their individual tastes.

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6. Conclusions and Future Work

6.1. Introduction

In this section, the conclusions drawn from the development and implementation of the project are summarized, along with a discussion of the key accomplishments and challenges faced throughout its lifecycle. Furthermore, potential future improvements and enhancements to the application are explored, providing a roadmap for ongoing refinement and expansion. By examining the lessons learned and opportunities for growth, a comprehensive understanding of the project's impact and potential trajectory can be formed, setting the stage for continued innovation and success in the ever-evolving field of web applications and recommender systems.

6.2. Issues and Risks

One of the key issues the project faced was faults that could occur in the content-based recommendation system that was chosen for this application. Content-based recommender systems compute recommendations based on the highest similarity between items/users, which is determined by classifications made about the data.

This issue faced during the development of this system was the risk of users getting trapped in an information confinement area (25). In this situation, items that get rated highly early in the application's life cycle tend to be recommended the most often. As these items continue to be recommended and receive high ratings, their classification value metric increases, causing the output to become skewed.

To address this issue, it was necessary to create multiple testing accounts and have them rate a diverse set of tattoos randomly. This approach helped distribute the ratings and recommendations more evenly, reducing the impact of the information confinement area in the early stages of the application's development.

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One of the key issues faced during the development of the content-based recommendation system was ensuring its scalability, as the number of users and items in the system could grow significantly over time. Scalability is crucial to ensure the system can handle a larger amount of data and provide timely recommendations to users, without compromising on performance or accuracy.

During the development of the project, another key challenge faced was the correct implementation of individualized user ratings for tattoos, ensuring that each user could save and rate tattoos specific to their account. Initially, the logic for saving tattoo ratings was not correctly tied to the unique user IDs, which resulted in ratings being shared across different users or not being saved correctly.

To address this issue, the system was modified to associate each saved tattoo rating with the user's unique ID. This change involved updating the data structure for storing the ratings in the database, as well as adjusting the logic for saving and retrieving these ratings in the application. By associating ratings with user IDs, it became possible to store and manage separate sets of saved and rated tattoos for each user. This ensured that each user's experience was personalized and tailored to their preferences.

Once these changes were implemented, users could save and rate tattoos independently of one another, with their ratings being securely stored and linked to their respective accounts. This enhancement greatly improved the overall user experience and allowed for a more accurate and personalized recommendation system that took into account the individual preferences of each user.

6.3. Plans and Future work

As with any software development project, there is always room for improvement and future enhancements. Many considerations could be made when planning future development and implementation of new technologies and functionalities for this project,

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however one of the most valuable functionalities that could be implemented would be the addition of a google maps API which could host a map, based on the users location, in the webapp, to display to the user the nearest surrounding tattoo shops where they could bring their recommendations to be tattooed.

As the recommendation system in the project grows, incorporating better scalability would be essential to maintain performance and provide accurate recommendations to a larger user base. To achieve improved scalability, the project could leverage distributed computing and implement a more efficient recommendation algorithm, such as matrix factorization or deep learning-based models. These techniques can significantly reduce the time complexity of the recommendation process, enabling the system to handle a larger volume of user data without compromising performance.

Additionally, the project could benefit from implementing a more robust database infrastructure that can handle increased traffic and scale horizontally as needed. Utilizing a scalable cloud-based database service, such as Google Cloud Firestore or Amazon DynamoDB, can provide the required flexibility and performance improvements for a growing user base.

By considering these approaches, the project would be better prepared to handle the increased demands of a growing recommendation system while maintaining a high level of user satisfaction and accurate, personalized recommendations. Another key area of future work which could be implemented would be developing a mobile application, although the webapp version has also been made to be responsive to multiple devices a lot of users opt to not use web applications on mobile, this in turn means creating a mobile version would help the tattoo web application to reach a much broader audience.

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6.3.1. GANTT Chart

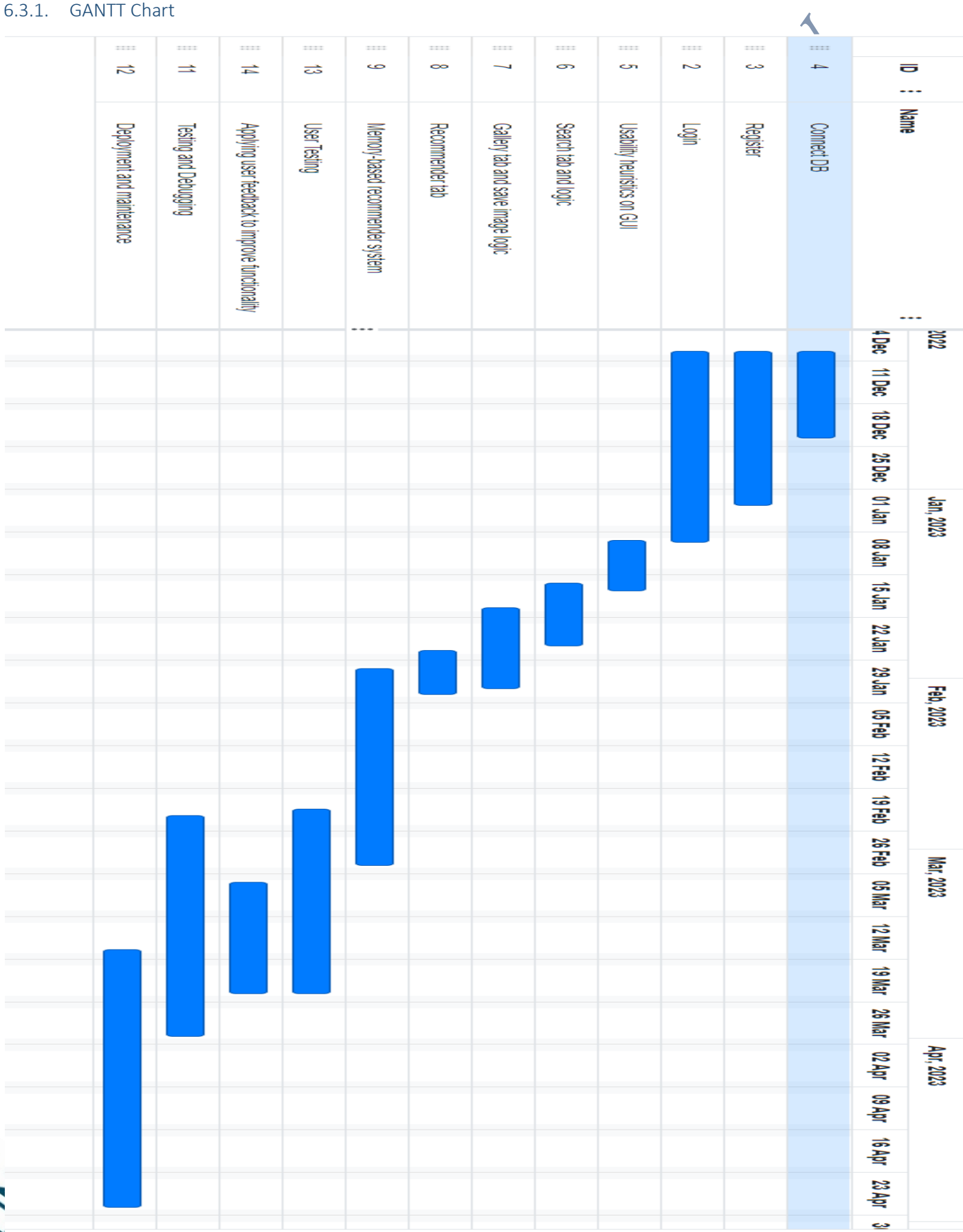

Figure 25 - GANTT Chart

6.4 Conclusion

In conclusion, the development of the tattoo web application involved a combination of front-end design, back-end implementation, recommendation algorithm development, and integration of all components. Throughout the development process, various challenges were encountered and addressed, resulting in a well-functioning and user-friendly platform.

React.js played a vital role in creating an intuitive and responsive user interface, while Firebase provided a scalable and real-time back-end solution. The custom recommendation algorithm allowed for personalized tattoo suggestions based on user preferences, enhancing the overall user experience.

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